

Manuscript submitted to the Journal of Financial Stability

This Time Is Not So Different: Volatility Regimes In The Cryptocurrency Market

Aliro Sanhueza^{a,*}

Nicolás Magner^a

^a School of Business and Economics, Universidad Diego Portales,
Avenida Santa Clara 979, Santiago, Chile

* Corresponding author. Email: aliro.sanhueza@mail.udp.cl

Date of submission: May 8, 2026

Abstract

In conventional markets, volatility admits a decomposition into a systematic component and an idiosyncratic residual. No such decomposition yet exists for a contemporary cryptocurrency universe at the individual-asset level. We build it by applying to a panel of 38 cryptoassets (91% of market capitalization, October 2018 to February 2026) the asymptotic principal components factor model of Connor, Korajczyk and Linton (2006) and the residual COVOL framework of Engle and Campos-Martins (2023), and we obtain a daily per-asset decomposition into common and idiosyncratic components. The cross-section alternates between two regimes that a Markov-switching regression separates sharply: systemic stress with a common share above 55%, and speculative below 30%. Within that architecture, the two largest single-day shocks in the sample do not occur during the crypto-native failures, but at the SEC v. Ripple ruling and the spot Bitcoin ETF approval; LUNA / Terra is the only episode with sustained elevation. Macro-financial transmission turns out to be selective: crypto-native sentiment, captured by the Fear & Greed index, reaches the idiosyncratic volatility of nearly the entire cross-section, while the VIX, the EPU, the NFCI, the GPR, the dollar and oil add no explanatory power at the daily frequency. Applied to 38 S&P 500 stocks and 11 SPDR sector ETFs under the same procedure, the exercise yields an indistinguishable regime architecture; the differential concentrates in the upper tail and in crypto-native events. In the space of volatility, this time is not so different.

Keywords: Cryptocurrencies; Co-volatility; COVOL; Factor model; GARCH

JEL classification: G12, G15, G17, G28

Author information and contributions

Aliro Sanhueza (corresponding author).

School of Business and Economics, Universidad Diego Portales,
Avenida Santa Clara 979, Santiago, Chile.

Email: aliro.sanhueza@mail.udp.cl.

Nicolás Magner.

School of Business and Economics, Universidad Diego Portales,
Avenida Santa Clara 979, Santiago, Chile.

Author contributions (CRediT taxonomy). Both authors contributed to: conceptualisation, methodology, formal analysis, software, validation, data curation, writing (original draft), writing (review and editing). Both authors read and approved the final manuscript.

Declarations

Funding. The authors declare that no specific funding was received for this work.

Competing interests. The authors have no competing interests, financial or otherwise, to disclose in relation to the content of this manuscript.

Data availability. All data used in this study are obtained from publicly available sources: cryptocurrency log-returns from CoinMarketCap; the VIX, the U.S. Dollar Index and WTI crude prices from the Federal Reserve Bank of St. Louis (FRED); the Economic Policy Uncertainty (EPU) index from policyuncertainty.com; the Geopolitical Risk (GPR) index from matteoiacoviello.com; the National Financial Conditions Index (NFCI) from the Federal Reserve Bank of Chicago; the Fear & Greed index from alternative.me. The full universe with tickers and market shares is reported in the Online Appendix.

Code availability. Replication code (Python) for the full APC + COVOL pipeline, the Markov-switching identification, the bootstrap inference and all tables and figures is available from the corresponding author upon reasonable request.

Ethical approval. Not applicable; the study uses only publicly available financial market data and does not involve human or animal subjects.

Use of generative AI. The authors used generative AI tools to assist with language editing of selected passages of the manuscript. The authors take full responsibility for the content of the publication.

Acknowledgements

The authors thank seminar participants at Universidad Diego Portales for helpful comments on earlier drafts of this work.

This Time Is Not So Different: Volatility Regimes In The Cryptocurrency Market

Abstract

In conventional markets, volatility admits a decomposition into a systematic component and an idiosyncratic residual. No such decomposition yet exists for a contemporary cryptocurrency universe at the individual-asset level. We build it by applying to a panel of 38 cryptoassets (91% of market capitalization, October 2018 to February 2026) the asymptotic principal components factor model of Connor et al. (2006) and the residual COVOL framework of Engle & Campos-Martins (2023), and we obtain a daily per-asset decomposition into common and idiosyncratic components. The cross-section alternates between two regimes that a Markov-switching regression separates sharply: systemic stress with a common share above 55%, and speculative below 30%. Within that architecture, the two largest single-day shocks in the sample do not occur during the crypto-native failures, but at the SEC v. Ripple ruling and the spot Bitcoin ETF approval; LUNA / Terra is the only episode with sustained elevation. Macro-financial transmission turns out to be selective: crypto-native sentiment, captured by the Fear & Greed index, reaches the idiosyncratic volatility of nearly the entire cross-section, while the VIX, the EPU, the NFCI, the GPR, the dollar and oil add no explanatory power at the daily frequency. Applied to 38 S&P 500 stocks and 11 SPDR sector ETFs under the same procedure, the exercise yields an indistinguishable regime architecture; the differential concentrates in the upper tail and in crypto-native events. In the space of volatility, this time is not so different.

Keywords: Cryptocurrencies, Co-volatility, COVOL, Factor model, Idiosyncratic risk

1. Introduction

One principle organises most of modern empirical asset pricing: the volatility of returns decomposes into a systematic component, captured by a small number of common factors, and an asset-specific residual. Market shocks are interpreted, portfolios are built and risk models are calibrated on top of that decomposition. The factor literature (Connor et al., 2006; Bai & Ng, 2002) has extended this apparatus from the first to the second moment, and Engle & Campos-Martins (2023) have recently developed a tool to identify, within the residuals of a factor model, the days on which the system experiences a common volatility shock. This logic organises the empirical study of equity, bond and currency markets today. It does not organise, however, the study of the cryptocurrency market. Despite three complete boom-and-bust cycles since the launch of Bitcoin, the literature has yet to deliver a per-asset decomposition of crypto volatility into a common cross-sectional component and an idiosyncratic residual for a broad and contemporary universe.

This omission is not marginal. Crypto market capitalization exceeds that of several G7 economies, its daily turnover rivals that of the major currency pairs, and institutionalisation through stablecoins and listed vehicles has tied its volatility to traditional markets (Galati & Capalbo, 2024; Xu et al., 2023). The joint failure of Silicon Valley Bank and the USDC stablecoin in March 2023 showed that a crypto episode can reach traditional banking within hours (Liu

et al., 2023). Yet the most basic risk-management question in this market, namely what fraction of each cryptoasset's volatility is shared with the rest of the system and what fraction is its own, currently admits no empirical answer. The consequences are concrete. Claims about intra-crypto diversification remain untested; the systemic footprint of regulatory action remains unquantified; and the rich factor literature on the returns side (Liu & Tsyvinski, 2021; Liu et al., 2022; Biais et al., 2023) has yet to find its counterpart on the volatility side.

The literature on crypto volatility has advanced along three parallel fronts. The first is the study of volatility at the individual-asset level: the GARCH family has documented the clustering, the fat tails and the persistence that characterise Bitcoin and the leading altcoins, and has established a technical benchmark on the univariate behaviour of each series (Dyhrberg, 2016; Katsiampa, 2017). The second is the cross-section of returns: factor models have shown that crypto valuation admits a familiar reading along Fama-French lines, and have disciplined the question of which factors explain the risk premium (Liu & Tsyvinski, 2021; Liu et al., 2022; Biais et al., 2023). The third is event studies: the LUNA / Terra collapse (Briola et al., 2023; Aramonte et al., 2021), the FTX bankruptcy (Conlon et al., 2024; Akyildirim et al., 2024) and the USDC depeg (Galati & Capalbo, 2024; Liu et al., 2023) have been analysed episode by episode, through prices, returns and simple contagion measures. The COVOL framework of Engle & Campos-Martins (2023) provides, on this map, a natural instrument for examining common volatility once the return factor has been removed; Pham et al. (2025) and Pham et al. (2026) have applied it to cryptocurrencies at the sectoral level and under directional asymmetry. None of these advances, however, has incorporated the recent phase of the market, marked by Bitcoin's consolidation as an institutional vehicle after the 2024 spot ETF approval, nor has any extended the exercise to the individual-asset level on a contemporary universe spanning the regulatory and institutional events of 2023 and 2024.¹ That phase also shifts the horizon of the question. As cryptocurrencies move from a sideline market into the traditional financial infrastructure, through stablecoins held in bank accounts, ETFs traded on regulated exchanges and judicial rulings that classify individual assets as securities, empirically assessing whether their volatility obeys the same regularities as conventional markets becomes a concrete question for investors who build portfolios, for risk managers who calibrate exposure models and for regulators who weigh the systemic footprint of their interventions.

Hence the question that motivates this paper: are the volatility dynamics of cryptocurrencies governed by the same regularities as conventional financial markets, or do they constitute, as much of the popular narrative on the asset class suggests, a genuinely *sui generis* structure? The analysis is built on a panel of 38 cryptoassets observed daily between October 2018 and February 2026. We first estimate an asymptotic principal components factor model following Connor et al. (2006), implemented through the iterative algorithm of Jones (2001), with the number of latent factors selected by the information criterion of Bai & Ng (2002). We then apply, on the residuals of the factor model, the residual COVOL framework of Engle & Campos-Martins (2023). The procedure delivers, for each asset and each day, a decomposition of total volatility into its common and idiosyncratic components, together with two scalar aggregates that summarise the state of the system. To anchor the comparative dimension of the question, we replicate the same procedure on two non-crypto benchmarks: a panel of 38 individual large-cap stocks from the S&P 500 and a panel of 11 SPDR sector ETFs.

¹Specifically: the USDC stablecoin depeg during the Silicon Valley Bank collapse (March 2023); the SEC v. Ripple ruling, which held that programmatic sales of XRP did not constitute the sale of a security (July 2023); the anticipation rally and subsequent approval of the first eleven spot Bitcoin exchange-traded funds (December 2023 to January 2024); the fourth Bitcoin halving (April 2024); and the post-election rally of late 2024.

The results accumulate evidence in the same direction. The crypto cross-section alternates between two clearly distinct regimes: in the systemic-stress regime the common factor absorbs more than 55% of the cross-sectional variance and every cryptoasset moves with the same shock, whereas in the speculative regime the common factor attenuates and asset-specific dynamics dominate. An endogenous Markov-switching identification (Hamilton, 1989) corroborates the architecture without imposing the descriptive thresholds. Within that architecture, the two largest single-day cross-sectional shocks in the entire sample are recorded not during the most heavily covered crypto-native failures, but at the SEC v. Ripple ruling (July 2023) and the spot Bitcoin ETF approval (January 2024); the LUNA / Terra collapse stands as the only episode in the period with sustained elevation of common volatility over a multi-week window.

At the individual-asset level, Bitcoin sits at the lower extreme of the crypto volatility distribution in absolute terms and retains that position in total volatility after controlling for capitalization; in the separate components, a non-trivial portion of the gap reflects the size effect. Macro-financial transmission at the daily frequency turns out, in turn, to be markedly selective: crypto-native sentiment, captured by the Fear & Greed index, transmits to the idiosyncratic volatility of nearly the entire cross-section, while the VIX, the EPU of Baker et al. (2016), the GPR of Caldara & Iacoviello (2022), the NFCI, the dollar and oil add no incremental explanatory power. Applied under the same procedure to the equity benchmarks, the crypto apparatus yields a regime architecture essentially indistinguishable from the conventional one, and the differential concentrates in the magnitude of crypto-native shocks in the upper tail. In sum, cryptocurrencies behave, in the essentials, like traditional markets; differences exist, but are bounded and appear almost exclusively in the upper tail of stress, during crypto-native events such as the LUNA collapse, the USDC depeg or the Bitcoin ETF rally.

The paper contributes to four strands of the literature. The cross-sectional factor evidence of Liu et al. (2022) on crypto returns finds its counterpart here on volatility, with the additional observation that the factor structure relevant for risk management is dynamic rather than time-invariant. The univariate GARCH evidence (Dyhrberg, 2016; Katsiampa, 2017) is complemented by an explicit decomposition that separates the common from the asset-specific. The literature on stress episodes (Briola et al., 2023; Conlon et al., 2024; Galati & Capalbo, 2024), focused until now on stablecoin and exchange failures, is enriched with the documentation of single-day regulatory and institutional shocks. And the existing COVOL literature (Engle & Campos-Martins, 2023; Campos-Martins & Hendry, 2024; Pham et al., 2025, 2026) is extended from the sectoral to the individual-asset level on a contemporary crypto universe. The implications cut across investment, regulation and risk management. For investors, intra-crypto diversification is state-contingent and is lost during systemic episodes. For regulators, interventions presented as narrow can produce single-day footprints that no crypto-native episode in the period produced. For risk managers, the insulation of crypto volatility from the VIX and from conventional macro variables at the daily frequency implies that crypto risk budgets cannot be built from the usual proxies.

The remainder of the paper is organised as follows. Section 2 places the paper within the relevant literatures. Section 3 describes the data and the empirical strategy. Section 4 reports the results. Section 5 concludes.

2. Literature review

One idea organises much of the empirical analysis of volatility in conventional markets: cross-sectional variance admits a decomposition into a common component, captured by a small number of latent factors, and an asset-specific residual (Herskovic et al., 2016). Jones (2001) implemented it

for the first time on heteroscedastic returns, through an iterative algorithm that alternates between cross-sectional principal components estimation and temporal rescaling of the residuals. Connor et al. (2006) formalised the asymptotic estimator and established the joint identification of the common and asset-specific components. The complementary problem, the selection of the number of factors in approximate models, was solved by Bai & Ng (2002), whose information criterion remains the standard tool; Barigozzi & Hallin (2017) subsequently extended the apparatus to generalised dynamic models with explicit volatility dynamics. On this basis, Engle & Campos-Martins (2023) built the COVOL framework: a multiplicative scalar that acts on the standardised cross-sectional shocks and identifies the days on which the cross-section experiences a common volatility event. Campos-Martins & Hendry (2024) carried the framework to the global carbon transition. Combined with a previously estimated factor model, this machinery delivers a daily per-asset decomposition of total volatility into common and idiosyncratic components, exactly the kind of decomposition that organises conventional asset pricing.

Carrying that logic to the crypto market was a natural extension. The crypto volatility literature, however, has advanced along parallel fronts without converging on that exercise. The GARCH family started by Engle (1982) and generalised by Bollerslev (1986) has been applied extensively to Bitcoin and to the largest cryptoassets (Dyhrberg, 2016; Katsiampa, 2017), and complemented by a realised-volatility branch that exploits high-frequency data for forecasting purposes (Bergsli et al., 2022). The univariate consensus is robust: crypto returns exhibit strong clustering, fat tails and substantial persistence relative to traditional asset classes. By construction, however, that consensus leaves the cross-sectional dimension uncharacterised: how much of an asset's volatility is shared with the rest of the cross-section and how much is its own (Zhang & Li, 2020). Pham et al. (2025) and Pham et al. (2026) carried the COVOL framework into the crypto space at the sectoral level and under directional asymmetry (Zhang & Zhao, 2023), but not at the individual-asset level.

The cross-section of crypto *returns*, by contrast, has been subjected to systematic factor analysis. Liu & Tsyvinski (2021) documented that crypto risk premia are economically and statistically significant, and that returns can be predicted with simple factors such as past returns and attention. Liu et al. (2022) extended the evidence to a three-factor model, namely market, size and momentum, in direct analogy with Fama-French (Fama & French, 1992). Biais et al. (2023) provided the theoretical counterpart: an equilibrium that rationalises the high crypto volatility from speculative demand, transactional demand and limited storage capacity. The natural question this literature poses, without answering it, is whether the factor structure that organises crypto returns also organises their volatility, and whether that organisation is stable along the cycle or varies with the state of the market. Returns-based models treat factor exposures as time-invariant, so the question does not admit a direct answer from their apparatus. Xu et al. (2023), in a complementary direction, documented the connectedness between the aggregate global COVOL and the major international volatility indices, including the VIX, but their analysis sits at the aggregate level and does not characterise the internal cross-section of the crypto world.

To this open question are added three substantive literatures that intersect, at this point, with the dynamics of crypto volatility. The first examines macro-financial transmission. Demir et al. (2018) found that the EPU of Baker et al. (2016) predicts Bitcoin returns; Fang et al. (2019) documented that global economic uncertainty matters for Bitcoin volatility and for its hedging effectiveness; Bouri et al. (2018) reported substantial spillovers between Bitcoin and other assets during bear markets; the GPR of Caldara & Iacoviello (2022) has been used with mixed results. This evidence has been built largely on aggregate Bitcoin or on small cross-sections, without separating the systemic from the idiosyncratic channel, a distinction that matters theoretically because each channel loads on a distinct component of volatility. The second literature catalogues

the stress episodes that the crypto market has experienced since 2018: the COVID-19 selloff of March 2020, the Chinese mining ban of mid-2021, the LUNA / Terra collapse of May 2022 (Briola et al., 2023; Aramonte et al., 2021), the FTX bankruptcy of November 2022 (Conlon et al., 2024; Akyildirim et al., 2024), the USDC depeg associated with the Silicon Valley Bank collapse of March 2023 (Galati & Capalbo, 2024; Liu et al., 2023), the SEC v. Ripple ruling of July 2023 and the spot Bitcoin ETF rally of late 2023. The literature has treated them in isolation through event studies on prices, returns or simple contagion measures (Briola et al., 2023; Conlon et al., 2024; Galati & Capalbo, 2024; Härdle et al., 2020), which leaves open the question of whether different episodes belong to the same empirical category: an algorithmic collapse, a regulatory ruling on an individual asset, and the approval of an institutional investment vehicle may, in principle, leave very different footprints on the volatility cross-section. The third literature debates Bitcoin's role. Baur et al. (2018) ask whether Bitcoin is a medium of exchange or a speculative asset and conclude that it behaves predominantly as the latter; Dyhrberg (2016) finds it has properties intermediate between gold and the dollar; Urquhart (2016) documents early informational inefficiencies; Härdle et al. (2020) provide a comprehensive overview that emphasises the heterogeneity of the cross-section. It is a debate conducted largely in qualitative terms, and it leaves unanswered an underlying quantitative question: what position does Bitcoin occupy within the crypto volatility distribution, and does that position reflect a structural property or the mechanical effect of its size?

A fourth literature, equally relevant, connects directly with the empirical exercise. Attention and sentiment are established determinants of financial volatility (Da et al., 2011); in the crypto world, Smales (2022) documented that investor attention, measured with Google searches and related proxies, predicts returns and volatility, and the Fear & Greed index synthesises several of these proxies into a daily indicator that has become an industry reference. The open question concerns the channel through which crypto sentiment operates: if retail sentiment moves all crypto-assets simultaneously, it should load on common volatility and operate as a systemic factor; if instead it translates into asset-specific dynamics, it should load on the idiosyncratic component and operate as a source of dispersion. Without a prior decomposition, the two readings are confounded.

The four substantive literatures thus converge on the same pending apparatus. The open questions in each of them, namely whether macro shocks operate through a systemic or an idiosyncratic channel, whether different stress episodes belong to the same empirical category, whether Bitcoin's special position is structural or mechanical, and whether sentiment operates on the common factor or on idiosyncratic dispersion, are all answered on a per-asset decomposition of crypto volatility into its common and idiosyncratic components. That decomposition is precisely what the factor apparatus of Connor et al. (2006), Jones (2001) and Bai & Ng (2002), combined with the COVOL framework of Engle & Campos-Martins (2023), makes it possible to construct, and which Section 3 develops in the context of a contemporary crypto universe. Taken as a whole, the literature also reveals a recurring tension between the popular framing of cryptocurrencies as an exotic asset class and the empirical evidence that, where the data have been examined carefully, they exhibit familiar regularities of conventional markets: factor structure in returns, state-dependent comovement, sensitivity to financial conditions and to sentiment, and a hierarchy of stress in which institutional events mix with the purely technological ones. That tension motivates the empirical question of the paper and guides the design of the exercises that follow.

3. Methodology

The paper's question requires, before any empirical exercise, separating what the crypto cross-section shares from what is specific to each asset. The factor literature provides the natural

apparatus for that separation; the COVOL framework provides the device that captures common volatility once the systematic component of returns has been removed. The methodological strategy that follows applies, in combination, the asymptotic principal components framework of Connor et al. (2006) and the COVOL setting of Engle & Campos-Martins (2023), on a daily panel of cryptoassets sufficiently broad and contemporary to support the exercise. The output is a parsimonious and time-varying decomposition of crypto volatility into its common (factor), idiosyncratic and total components, together with two scalar aggregates that summarise the state of the system on each day. The remainder of this section describes, in order, the data, the estimation procedure and the two empirical exercises that the decomposition enables.

3.1. Data

The panel must be wide to identify a latent factor structure, long enough to span at least one full crypto cycle, and contemporary enough to include the regulatory and institutional events of 2023 and 2024. It comprises daily log-returns of 38 cryptoassets between 9 October 2018 and 6 February 2026, for a total of 1,908 trading days per asset. The starting date is fixed by the joint availability of high-quality price data: prior to October 2018, several altcoins had insufficient volume or were not listed on exchanges, conditions that would compromise the estimation of a stable factor structure. The universe is built by intersecting the largest cryptoassets at the end of the sample with the requirement of continuous data over the entire window. The 38 retained cryptoassets cover about 90.92% of total market capitalization as of December 2025: Bitcoin and Ethereum account for 68.5%, USDT and USDC for another 11.2%, and the remaining 34 cryptoassets for the balance.² For each asset i and each day t we compute the log-return $r_{i,t} = \ln(p_{i,t}/p_{i,t-1})$. For the cross-sectional analysis, the universe is sorted into four capitalization tiers: four mega-caps (Bitcoin, Ethereum, BNB, XRP); ten large-caps (Cardano, Dogecoin, TRON, Chainlink, Bitcoin Cash, Litecoin, Stellar, Filecoin, Monero, Ethereum Classic); 22 mid-caps spanning DeFi tokens, smart-contract platforms, privacy coins and gaming or metaverse assets; and the two leading stablecoins (USDT, USDC). The stablecoins are retained because their small but non-zero volatility carries information about the factor structure.

Conditioning the sample on the existence of continuous data through to the end of the window introduces survivorship bias, and it is worth making its implications explicit before moving on. Three economically prominent cryptoassets with terminal failures during the period (Terra / LUNA and TerraUSD in May 2022, and FTT in November 2022) fall outside the panel. The consequences are twofold. During the LUNA / Terra collapse and the FTX bankruptcy, the decomposition captures cross-sectional contagion onto the surviving assets, not the variance of the failed tokens: the magnitudes during those windows are therefore upper bounds on the systemic content, since the absence of the most volatile assets mechanically inflates the share of variance attributable to the common factor. And, more generally, conditioning on survival removes the assets with the highest idiosyncratic volatility, biases the distribution downward and probably amplifies Bitcoin’s apparent role as an anchor in the pairwise exercises of Section 4.3. The first consequence is addressed directly in Section 4.5 (Appendix Table 8), by reaggregating the cross-section with LUNA, TerraUSD and FTT included up to their last trading dates: the qualitative regime classification is preserved, although the common share during the FTX window falls by 14 percentage points. A more complete extension would be a dynamically rebalanced universe in the spirit of Liu et al. (2022),

²Market shares are taken from CoinMarketCap as of December 2025. Online Appendix Table 9 reports the complete listing with tickers and market shares; the per-asset descriptive statistics of log-returns are in Online Appendix Table 8.

which is discussed as a limitation in Section 5. It is worth noting, in any case, that the 38 retained cryptoassets capture most of the aggregate market capitalization and that the loss of representativeness concentrates in the long tail of small tokens: the bias operates on the *volatility distribution*, not on the coverage of market value.

3.2. Volatility decomposition: an APC factor model with GARCH dynamics

The decomposition is built on this panel. Let $\tilde{r}_{i,t}$ denote the standardised log-return of asset i at time t , with $i = 1, \dots, N$ and $t = 1, \dots, T$. We posit an approximate K -factor structure

$$\tilde{r}_{i,t} = \sum_{j=1}^K \lambda_{i,j} f_{j,t} + e_{i,t}, \quad (1)$$

where $f_{j,t}$ is the j -th latent common factor, $\lambda_{i,j}$ is the loading of asset i on factor j , and $e_{i,t}$ is the idiosyncratic disturbance. Loadings and scores are estimated jointly through the iterative algorithm of Jones (2001), which alternates between a cross-sectional regression that updates the factors and a time-series regression that updates the loadings until convergence.

The number of factors is selected following the criterion of Bai & Ng (2002), which minimises

$$\text{IC}(K) = \ln V(K, \hat{F}^K) + K \cdot g(N, T), \quad (2)$$

where $V(K, \hat{F}^K)$ is the average residual variance after extracting K factors and $g(N, T)$ is a penalty that grows with the panel dimension. Table 1 reports the three variants of $g(N, T)$ proposed by Bai & Ng (2002) for $K = 0, 1, \dots, 8$. The three criteria reach their minimum at $K = 2$. The marginal gain from a third factor is small: 2.6 percentage points of the cross-sectional variance, against the 56.2 of the first and the 4.1 of the second. We adopt $K = 2$ as the main specification and report the decomposition under $K = 3$ in Appendix 2.1 as a robustness check.³

The factor structure admits a direct economic reading. The first factor captures the broad crypto market: every non-stablecoin cryptoasset loads positively on it, with loadings in a narrow band between 0.10 and 0.20, and USDT and USDC load close to zero. We interpret it as an aggregate crypto-market factor that captures the risk-on and risk-off shocks that move every unanchored asset simultaneously. The second factor is, by construction, a stablecoin or fiat-anchor factor: it absorbs the dimension that separates the anchored from the unanchored, with USDT and USDC loading around 0.70 and the rest of the cross-section close to zero.⁴ The first factor accounts for 56.2% of the cross-sectional variance and the second adds another 4.1%; the remaining 39.7% is asset-specific variation, which the GARCH machinery described below absorbs at the individual level.

The economic content concentrates, therefore, on the first factor. It is worth asking whether it is homogeneous or whether it admits internal structure across capitalization tiers. The evidence is one of homogeneity: the interquartile range of the factor-1 loadings is [0.16, 0.18] across the 36 non-stablecoin cryptoassets, with no systematic differences between mega-caps (mean 0.175), large-caps (0.160) and mid-caps (0.166); the loading on Bitcoin (0.181) is indistinguishable from

³The scree plot in Appendix 3 shows that the first factor concentrates most of the comovement; the second adds a clearly separable dimension (the stablecoin block); subsequent components flatten out.

⁴Since the second factor operates as a stablecoin dummy, a single-factor specification on the 36 non-stablecoin cryptoassets is empirically equivalent to the $K = 2$ one on the full panel. Online Appendix Table 2 reports this robustness; the within-episode market share is preserved within tenths of a percentage point under either choice.

Table 1: Bai and Ng (2002) information criteria for the number of factors.

K	$\ln V(K)$	$IC_1(K)$	$IC_2(K)$	$IC_3(K)$
0	-0.0005	-0.0005	-0.0005	-0.0005
1	-0.8259	-0.7288	-0.7283	-0.7302
2	-0.9254	-0.7312	-0.7301	-0.7339
3	-0.9930	-0.7017	-0.7001	-0.7058
4	-1.0550	-0.6666	-0.6645	-0.6721
5	-1.1189	-0.6334	-0.6308	-0.6403
6	-1.1833	-0.6007	-0.5975	-0.6089
7	-1.2453	-0.5656	-0.5619	-0.5753
8	-1.3076	-0.5308	-0.5265	-0.5418

Note: Bai & Ng (2002) information criteria for selecting the number of factors K in an approximate factor model. $V(K)$ is the average residual variance after extracting K principal components from the panel of standardised log-returns ($T = 1908$, $N = 38$). Each criterion has the generic form $IC(K) = \ln V(K) + K \cdot g(N, T)$ with a different penalty $g(N, T)$. Bold entries mark the minimiser of IC_2 , the most commonly reported specification. The three criteria all select $K = 2$, the specification adopted in the main analysis. The alternative with $K = 3$ is reported in Appendix 2.1 as a robustness check.

that on Ethereum (0.192) and from the cross-sectional median. The heterogeneity that Section 4.2 documents at the per-asset volatility level emerges, then, from the time-varying variance of the common factor and from idiosyncratic dynamics, not from differential factor exposures.

The model in Equation 1 delivers, for each asset, a factor-driven component $\sum_j \lambda_{i,j} f_{j,t}$ and an idiosyncratic residual $e_{i,t}$. On each of them we fit a univariate sGARCH(1,1) with Student- t innovations to recover the conditional standard deviations. We denote them $\sigma_{i,t}^c$ and $\sigma_{i,t}^{\text{id}}$. Since both components are orthogonal by construction, the total volatility of asset i at t is

$$\sigma_{i,t}^T = \sqrt{(\sigma_{i,t}^c)^2 + (\sigma_{i,t}^{\text{id}})^2}. \quad (3)$$

The three per-asset series $\sigma_{i,t}^c$, $\sigma_{i,t}^{\text{id}}$ and $\sigma_{i,t}^T$ are the basic objects of the rest of the analysis. To make the relative weight of the components comparable across cryptoassets, we also report, for each i , the average common share $\bar{\sigma}_i^c / \bar{\sigma}_i^T$.

To summarise the COVOL state of the system on each day, we follow Engle & Campos-Martins (2023) and construct two scalar aggregates. The first, φ_t , captures the time-varying intensity of the common factor over the cross-section. The second, γ_t , captures the corresponding intensity of asset-specific shocks. Both are positive but not bounded in $[0, 1]$, so they do not admit a direct interpretation as variance proportions. To obtain a bounded quantity we define the daily market share of the common factor as

$$MS_t = \frac{\varphi_t}{\varphi_t + \gamma_t} \in [0, 1]. \quad (4)$$

MS_t measures the fraction of total cross-sectional variance attributable to the common factor on day t . Values near one indicate that the crypto universe collapses onto a single factor; values near zero, idiosyncratic dispersion in which each cryptoasset moves on its own narrative.

MS_t has a limitation: φ_t and γ_t scale with the absolute level of crypto volatility, and in episodes of high global volatility both can be simultaneously large, so that MS_t may take values comparable to those of low-volatility episodes even when the composition of risk has changed. As a check, we also compute the normalised market share

$$MS_t^* = \frac{\varphi_t/\bar{\varphi}}{\varphi_t/\bar{\varphi} + \gamma_t/\bar{\gamma}}, \quad (5)$$

where $\bar{\varphi}$ and $\bar{\gamma}$ are the long-run sample means. By rescaling each component by its historical level, MS_t^* is invariant to the absolute size of the volatility regime. The qualitative classification (systemic stress vs. speculative) is preserved under MS_t^* for the thirteen episodes of Section 4.1, so we report only MS_t in the main text.

A final clarification before the empirical analysis. Two aggregations of the common share appear throughout the paper. The first, used in the body and reported as $\overline{MS}$ in Table 2, is the within-window mean of the daily MS_t , computed from the COVOL scalars φ_t and γ_t . The second, used in some robustness exercises in Section 4.5, is the within-window cross-sectional ratio constructed on the GARCH-smoothed components: $MS^{xs} = \overline{\sigma^{c2}}/(\overline{\sigma^{c2}} + \overline{\sigma^{id2}})$. They differ for two reasons: MS_t uses the COVOL-ML scalars, which are heavy-tailed at the daily frequency, while MS^{xs} uses the smoothed components; and the order of aggregation differs (mean of ratios vs. ratio of means), which introduces a Jensen-inequality wedge. The full-sample mean of MS_t is 40.8%, while that of MS^{xs} is 59.7%. The qualitative regime classification is preserved under both aggregations; the rank correlation across the thirteen episodes is positive but moderate (Spearman $\rho = 0.50$). Throughout the paper we report which aggregation underlies each figure. We use MS_t for the main regime claims; we use MS^{xs} only when it provides cleaner robustness diagnostics, as in the leave-one-out and stablecoin-free exercises.⁵

3.3. Empirical specifications

With the decomposition in hand, the paper builds two empirical exercises that connect directly with the research question. The first examines the position of Bitcoin within the volatility cross-section. The motivation is twofold. The debate on Bitcoin's differential role has been conducted largely in qualitative terms and it is useful to translate it into a precise quantitative claim. And, more substantively, in conventional markets a reference asset that anchors the cross-section below the rest is one of the structural traits that portfolio theory and risk-management practice take for granted; verifying whether Bitcoin plays that role therefore offers a concrete way to assess whether cryptocurrencies reproduce the organisation of conventional markets on this point too. To make the question operational, we evaluate whether each cryptoasset is on average more or less volatile than Bitcoin in a component $k \in \{c, id, T\}$ from the daily differential

$$d_{j,t}^k = \sigma_{BTC,t}^k - \sigma_{j,t}^k, \quad (6)$$

for each asset $j \neq BTC$, and we regress it on a constant. The estimated constant $\bar{d}_j^k$ identifies the average daily distance in component k , and a negative value indicates that asset j is on average more volatile than Bitcoin in that component. Standard errors follow Newey & West (1987) with the standard truncation lag for daily series.⁶ To distinguish what is structural from what is

⁵Online Appendix Table 1 reports a side-by-side reconciliation of the two aggregations across the thirteen episodes.

⁶Specifically, $L = \lfloor 4(T/100)^{2/9} \rfloor = 7$.

mechanical, we complement the pairwise differentials with a cross-sectional regression of average per-asset volatility on the $\log_{10}$ market-cap share: this step makes it possible to contrast Bitcoin’s unconditional position with its residual position once size is controlled for.

The second exercise examines the determinants of each component’s dynamics. To make the question operational, we first need to fix the set of structural regressors. The selection is guided by the research question: if crypto volatility is governed by the same regularities as conventional markets, the variables that move conventional volatility should also move it here, while a distinctive crypto-native channel would have to operate through a sentiment indicator specific to the asset class. We therefore include the CBOE Volatility Index (*VIX*), the EPU index of Baker et al. (2016), the GPR index of Caldara & Iacoviello (2022), the Chicago Fed NFCI, the log-return of the U.S. Dollar Index (*DXY return*), the log-return of WTI crude (*Oil return*) and the crypto-native Fear & Greed index. All series are obtained from public sources and aligned to the daily frequency with the return panel.⁷ For each cryptoasset i and each component $k \in \{c, id\}$, we estimate

$$\sigma_{i,t}^k = \alpha_i^{(k)} + \sum_{\ell \in \{1, 5, 21\}} \rho_{i,\ell}^{(k)} \sigma_{i,t-\ell}^k + \sum_{j=1}^J \beta_{i,j}^{(k)} X_{j,t}^* + \varepsilon_{i,t}^{(k)}, \quad (7)$$

where $X_{j,t}^*$ is the j -th structural variable, standardised to zero mean and unit variance over the sample so that the coefficients $\beta_{i,j}^{(k)}$ are directly comparable in magnitude across regressors. Standard errors follow Newey-West with $L = 7$ lags. For each pair (i, k) , the null hypothesis $H_0 : \beta_{i,j}^{(k)} = 0$ tests whether $X_{j,t}^*$ adds explanatory power beyond the autoregressive structure of $\sigma_{i,t}^k$.

The choice of lags $\ell \in \{1, 5, 21\}$ rests on a prior diagnostic. For each asset and each component we fit pure $AR(p)$ models with $p \in \{1, 2, \dots, 30\}$ and select the order that minimises the AIC (Akaike, 1974). For the common component the AIC selects $p = 1$ uniformly, indicating that one lag is enough to capture its high persistence. For the idiosyncratic and total components the selected order ranges from 2 to 30, with a cross-sectional median of 5 days and an interquartile range of 4 to 7. That pattern motivates the parsimonious $\{1, 5, 21\}$ structure: the first lag captures the daily persistence of the common component, the fifth captures the weekly horizon at which the AIC concentrates for the other two, and the twenty-first adds a lower-frequency monthly anchor.⁸ We estimate the regression separately for each asset and each component, rather than pooling. Pooling would impose homogeneity on persistence and on macro sensitivity, dimensions on which the prior literature documents heterogeneity across cryptoassets (Liu & Tsyvinski, 2021; Biais et al., 2023); the per-asset strategy increases the number of regressions, but lets the data reveal whether the determinants are universal or asset-specific.

4. Results

The results are organised in five steps, from the aggregate market dynamics down to the per-asset detail. The first step (Section 4.1) places the crypto market against thirteen historical

⁷The VIX, the U.S. Dollar Index and the WTI crude price come from FRED (Federal Reserve Bank of St. Louis); the EPU is obtained from policyuncertainty.com, the repository maintained by the authors of the index; the GPR comes from the official site of Caldara & Iacoviello (2022) at matteoiacoviello.com; the NFCI is from the Federal Reserve Bank of Chicago; the Fear & Greed index is downloaded from the historical record published by alternative.me, the standard industry reference for the indicator.

⁸We re-estimate Equation 7 with the per-asset AIC-selected lag length and the qualitative conclusions do not change; the parsimonious $\{1, 5, 21\}$ structure has the additional advantage of making specifications directly comparable across assets.

episodes through the COVOL aggregates, and characterises the regime architecture that emerges from that exercise. The second (Section 4.2) disaggregates that aggregate picture into per-asset components and examines how volatility is distributed along the capitalization tiers. The third (Section 4.3) pins down Bitcoin’s position within that distribution and separates the structural from the mechanical through a residualisation by capitalization. The fourth (Section 4.4) identifies the macro-financial and crypto-native channels through which volatility is transmitted at the daily frequency. Section 4.5 closes with a battery of exercises that probe each identification choice and, above all, anchors the comparative dimension of the question against an equity benchmark estimated under the same procedure.

4.1. Volatility dynamics during key market episodes

The two scalars delivered by the COVOL decomposition, φ_t and γ_t , summarise on each day the cross-sectional variance attributable to the common factor and to the idiosyncratic residuals. Their relative weight, encoded in the market share MS_t of Equation 4, indicates whether the crypto cross-section moves together or apart. Values near one signal that all cryptoassets respond to the same shock; values near zero, idiosyncratic dispersion. Table 2 reports the behaviour of these aggregates over thirteen episodes covering the full sample: macroeconomic shocks, speculative phases, stablecoin and exchange failures, and major regulatory events. The episodes thereby become readable as points on a single mapping of events to volatility regimes.

What Table 2 reveals first is a sharp separation between two regimes. The systemic-stress episodes exhibit average market shares around or above 50%: the Chinese mining ban of May 2021 ($\overline{MS} = 51.3\%$), the LUNA / Terra collapse of May 2022 (58.4%), the USDC depeg during the SVB crisis of March 2023 (63.6%) and the FTX bankruptcy of November 2022 (46.5%). Their daily peaks exceed 90% and their φ_t medians are elevated relative to the sample reference. The crypto market collapses, in these moments, onto a single factor: asset-specific dispersion compresses and every cryptoasset moves with the latent common shock.

The opposite pattern appears in the speculative episodes. The DeFi summer of June to August 2020, the early-2021 altcoin season, the bull-run peak of March and April 2021 and the November 2021 all-time highs show market shares between 21.8% and 35.5%, with daily minima close to zero. The DeFi summer is the most extreme, with $\overline{MS} = 21.8\%$. In these periods the individual tokens experience explosive dynamics while the broad market factor remains relatively quiet.

A more subtle distinction appears within the stress regime itself, and it is worth reporting carefully because the decomposition picks up two qualitatively distinct phenomena that should not be confused. One is sustained elevation of the common volatility scalar over an event window. Among the thirteen episodes, only LUNA / Terra meets that condition: its within-window median is roughly three times the sample median, and its within-window mean remains 1.5 times the sample mean even after dropping the highest day (Appendix Table 7). The other is the occurrence of an extreme single-day cross-sectional shock embedded in an otherwise quiet window. The two largest values of φ_t in the entire sample are recorded at the SEC v. Ripple ruling (13 July 2023; $\varphi_t = 1.60$) and at the spot Bitcoin ETF approval (10 January 2024; $\varphi_t = 1.53$), each more than three orders of magnitude above the sample median of 0.0005. Outside those days, however, the SEC v. Ripple and BTC ETF windows are unremarkable: the within-window median of φ_t sits at the sample median or below, only two or three days exceed $\varphi_t > 0.01$, and the trimmed mean falls to 0.31 and 0.37 times the sample mean. The raw multipliers of 26 and 10 times that appear in the $\overline{\varphi}_t$ column of Table 2 should therefore be read as descriptions of where the largest daily shocks of the sample fall, not as evidence of sustained elevation over multi-week windows.

The balance, read together, is twofold. LUNA / Terra stands as the only episode in the period with sustained elevation of common volatility; the SEC v. Ripple ruling and the spot Bitcoin ETF approval contain the two largest single-day shocks, but the surrounding days are at sample levels. Section 4.5 backs this distinction with moving-block bootstrap confidence intervals (Appendix Table 10), robustness to symmetric windows (Appendix Table 11) and trimmed multipliers (Appendix Table 7). The literature has emphasised algorithmic stablecoin and exchange failures as the canonical episodes of crypto stress (Briola et al., 2023; Conlon et al., 2024; Galati & Capalbo, 2024); this analysis confirms LUNA / Terra in that role and, separately, identifies two single-day announcement events as the largest cross-sectional shocks of the period.

Figure 1 provides the visual counterpart to Table 2. The upper panel reports the seven-day centred moving averages of φ_t and γ_t on a logarithmic scale; the lower panel reports the daily market share, with reference lines at 30% and 55%. Five episodes are shaded for visual clarity, each illustrating a distinct regime. The COVID-19 crash of March 2020 produces the only joint peak in φ_t and γ_t in the entire sample; the market share remains in the 30–40% band, consistent with a hybrid macro shock. The altcoin season of January and February 2021 exhibits the opposite configuration: γ_t persistently dominates φ_t and the market share touches the 30% reference, signalling a speculative regime. The USDC depeg of March 2023 is the largest crypto-native stress event in the sample: φ_t jumps abruptly, γ_t stays compressed and the market share crosses 95% on at least one day. The SEC v. Ripple ruling of July 2023 produces the largest single-day φ_t value, with a market share of 100%; the late-2023 ETF anticipation rally produces the second, accompanied by a simultaneous rise in γ_t . Outside the shaded periods, the crypto-native failures of 2021–2022 (Chinese mining ban, LUNA / Terra, FTX) appear as pronounced but smaller spikes. The post-ETF segment shows a general decline in both scalars and a market share that stabilises around 40–50%, consistent with a calmer institutional regime.

Table 2: Aggregate COVOL dynamics during key cryptocurrency market episodes.

Episode	Period	N	$\bar{\varphi}_t$	$\tilde{\varphi}_t$	$\varphi_{\max}$	$N_{\varphi>0.01}$	$\bar{\gamma}_t$	MS	MS range
COVID crash	2020-02-15 – 2020-03-31	32	0.0107	0.0006	0.2530	4/32	0.0028	35.8%	[1.1%, 94.3%]
DeFi summer	2020-06-01 – 2020-08-31	66	0.0009	0.0004	0.0070	0/66	0.0023	21.8%	[0.2%, 83.6%]
Altcoin season	2021-01-01 – 2021-02-28	41	0.0043	0.0017	0.0240	5/41	0.0062	35.5%	[0.1%, 93.7%]
Bull-run peak	2021-03-01 – 2021-04-30	45	0.0020	0.0007	0.0070	0/45	0.0036	28.6%	[0.1%, 78.5%]
China mining ban	2021-05-15 – 2021-07-31	55	0.0060	0.0014	0.1610	5/55	0.0015	51.3%	[0.2%, 97.5%]
Nov 2021 ATH	2021-10-15 – 2021-11-30	33	0.0018	0.0004	0.0120	1/33	0.0024	31.9%	[0.7%, 93.9%]
LUNA / Terra	2022-05-01 – 2022-06-15	33	0.0056	0.0013	0.0490	5/33	0.0011	58.4%	[0.6%, 99.3%]
FTX collapse	2022-11-01 – 2022-12-15	33	0.0030	0.0003	0.0320	3/33	0.0007	46.5%	[1.4%, 91.0%]
USDC depeg / SVB	2023-03-08 – 2023-03-20	9	0.0093	0.0014	0.0600	2/9	0.0008	63.6%	[6.5%, 98.5%]
SEC v Ripple	2023-07-10 – 2023-07-31	16	0.1012	0.0001	1.6010	2/16	0.0012	37.2%	[1.8%, 100.0%]
BTC ETF anticipation	2023-12-01 – 2024-01-31	44	0.0361	0.0005	1.5260	3/44	0.0104	47.8%	[1.9%, 100.0%]
BTC halving 2024	2024-04-01 – 2024-05-15	33	0.0024	0.0003	0.0200	3/33	0.0007	46.4%	[0.5%, 97.2%]
Trump rally	2024-11-04 – 2024-12-15	30	0.0035	0.0011	0.0300	3/30	0.0022	46.5%	[0.4%, 92.2%]
Full sample	2018-10-09 – 2026-02-06	1,908	0.0038	0.0005	1.6010	83/1,908	0.0020	40.8%	[0.1%, 100.0%]

Note: Aggregate volatility statistics during thirteen historical episodes. $\bar{\varphi}$ and $\tilde{\varphi}$ denote, respectively, the mean and the median of the daily COVOL factor variance scalar over the episode window. $\varphi_{\max}$ is the largest single-day value; $N_{\varphi>0.01}$ is the count of trading days within the window with $\varphi_t > 0.01$, a threshold corresponding to roughly $20\times$ the full-sample median. Reporting both mean and median makes explicit that several high-impact episodes contain one or two extreme single-day spikes that dominate the mean. $\bar{\gamma}_t$ is the mean of the idiosyncratic variance scalar; MS is the mean of the daily common share, with the within-episode min and max in brackets. The last row reports the full-sample reference.

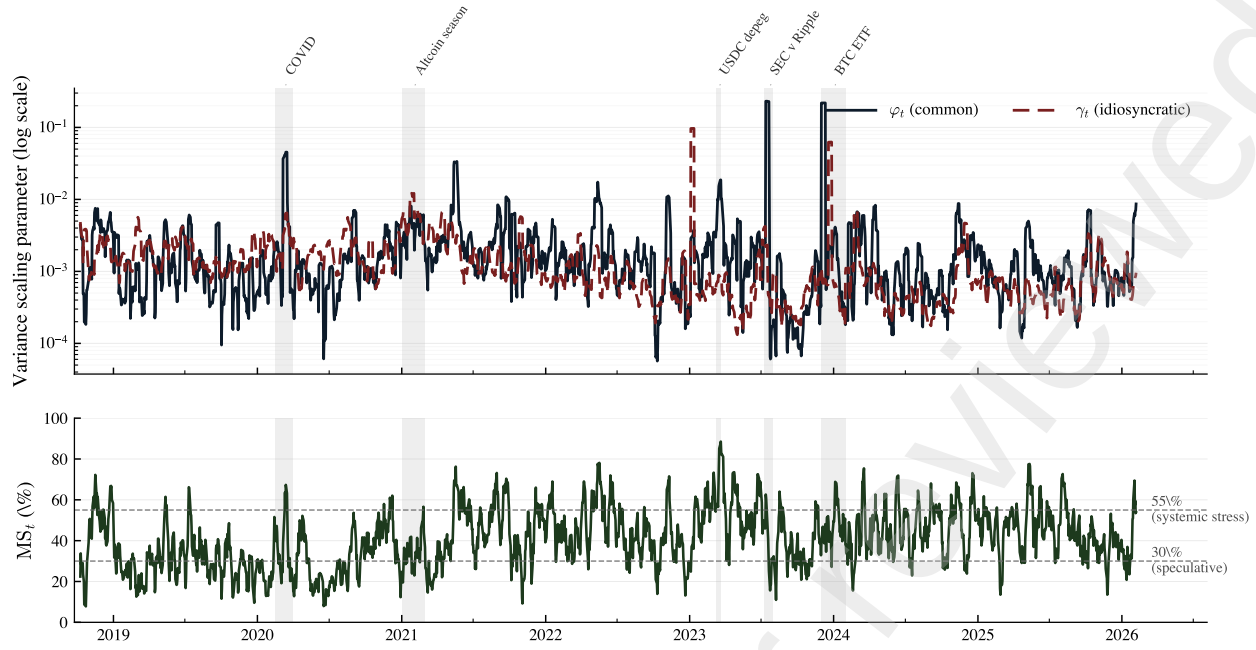


Figure 1: Daily evolution of the COVOL aggregates over the sample. Upper panel: common-variance scale parameter (φ_t , solid line) and idiosyncratic (γ_t , dashed line), seven-day centred moving averages, logarithmic scale. Lower panel: daily market share $MS_t = \varphi_t / (\varphi_t + \gamma_t)$ in percent (seven-day centred moving average), with horizontal reference lines at the 30% and 55% thresholds that approximate the boundaries between regimes documented in Table 2. The shaded vertical bands mark five highlighted episodes selected for visual clarity from the thirteen episodes of Table 2: the COVID-19 crash, the early-2021 altcoin season, the USDC depeg, the SEC v. Ripple ruling and the BTC ETF anticipation rally.

4.2. Descriptive statistics

The aggregate evidence in the previous subsection establishes that the crypto market alternates between two regimes. The immediate question is whether the cross-section that sits behind those aggregates is organised in a way that is recognisable to conventional asset pricing. To answer it, we disaggregate the aggregates into per-asset components. Table 3 reports a panel-level summary of the three estimated components $\sigma_{i,t}^c$, $\sigma_{i,t}^{\text{id}}$ and $\sigma_{i,t}^T$, computed within each capitalization tier; the full per-asset breakdown is in Online Appendix Table 6. The ordering by tier is not cosmetic: it captures a natural asset-pricing prior, namely that the ratio of systematic to idiosyncratic volatility should decrease as one moves from the most established assets to the smaller and more speculative ones.

Table 3: Volatility components by capitalization tier: panel-level summary.

Tier	N	$\bar{\sigma}^c$ (mean $\pm$ SD)	$\bar{\sigma}^{\text{id}}$ (mean $\pm$ SD)	$\bar{\sigma}^T$ (mean $\pm$ SD)	$\bar{\sigma}^c / \bar{\sigma}^T$
Mega-caps	4	0.032 $\pm$ 0.005	0.025 $\pm$ 0.005	0.043 $\pm$ 0.007	0.759
Large-caps	10	0.038 $\pm$ 0.004	0.032 $\pm$ 0.007	0.052 $\pm$ 0.006	0.724
Mid-caps	22	0.042 $\pm$ 0.004	0.034 $\pm$ 0.007	0.057 $\pm$ 0.004	0.743
Stablecoins	2	0.00023 $\pm$ 0.00013	0.00195 $\pm$ 0.00001	0.00200 $\pm$ 0.00004	0.116

Note: Cross-sectional means and standard deviations of the per-asset average volatility components, computed within each capitalization tier across the 38 cryptoassets. The last column reports the mean common share $\bar{\sigma}_i^c / \bar{\sigma}_i^T$ averaged within each tier. The full per-asset breakdown is reported in Online Appendix Table 6.

Four readings emerge from the table, and they reinforce one another. The absolute level of each component rises monotonically across the non-stablecoin tiers: average $\bar{\sigma}^c$ goes from 0.032 (mega-caps) to 0.038 (large-caps) and 0.042 (mid-caps), and $\bar{\sigma}^{\text{id}}$ exhibits a similar gradient. The common share $\bar{\sigma}^c/\bar{\sigma}^T$, by contrast, is broadly comparable across non-stablecoin tiers (0.72 to 0.76); the rising volatility level is therefore not a mechanical consequence of a decreasing factor exposure, but of an increase in absolute variance within the same factor structure. Stablecoins fall outside that pattern, with $\bar{\sigma}^c$ two orders of magnitude smaller than any non-stablecoin tier and a common share that collapses to 0.12: a figure that validates the factor structure by confirming that the broad factor barely loads on anchored assets. Finally, the cross-sectional dispersion within each tier is non-trivial, indicating that within-tier heterogeneity is part of the structure that the per-asset decomposition recovers, not residual noise.

4.3. Bitcoin's position in the volatility cross-section

The descriptive evidence suggests that Bitcoin sits at the lower extreme of the crypto volatility cross-section in absolute terms. The observation, however, admits two readings that are worth separating before formalising the exercise. A low unconditional position may be a mechanical effect of size: in any cross-section ordered by capitalization, the largest asset will tend to exhibit lower idiosyncratic volatility for diversification reasons that are not crypto-specific. A low conditional position, by contrast, is a structural property. We therefore examine the two dimensions separately. For the first, we estimate the constant $\bar{d}_j^k$ of the pairwise differential $d_{j,t}^k$ defined in Equation 6, separately for each component k and each asset $j \neq \text{BTC}$. Table 4 reports the results; a negative coefficient indicates that j is on average more volatile than Bitcoin in component k .

Table 4: Average daily volatility differentials against Bitcoin at the panel level.

Panel	N	Δ Common Vol.	Δ Idio. Vol.	Δ Total Vol.
Mega-caps	3	−0.009	−0.007	−0.012
Large-caps	10	−0.012	−0.013	−0.019
Mid-caps	22	−0.017	−0.014	−0.023
Stablecoins	2	+0.025	+0.018	+0.031

Note: Cross-sectional means within each tier of the per-asset constants $\bar{d}_j^k$, where $d_{j,t}^k = \sigma_{\text{BTC},t}^k - \sigma_{j,t}^k$ is the daily differential in component $k \in \{c, \text{id}, T\}$ between Bitcoin and asset j . A negative panel mean indicates that the cryptoassets in the panel are on average more volatile than Bitcoin in that component. Each per-asset $\bar{d}_j^k$ is significant at the 1% level under Newey-West HAC standard errors with $L = 7$ lags; the per-asset detail is reported in Online Appendix Table 7.

With $T = 1,908$ daily observations, small mean differentials can pass conventional significance thresholds without being economically large. We therefore complement the evidence with effect-size diagnostics that are robust to sample size. Appendix Table 9 reports the fraction of days on which Bitcoin volatility is strictly smaller than that of each non-stablecoin cryptoasset, by component, together with Cohen's d on the daily total-volatility differential. The average fraction reaches 0.99 for total volatility and Cohen's d averages 1.17, in the large-effect range. The unconditional finding is therefore robust.

It remains to examine the more substantive dimension, the size effect. Appendix Table 10 reports a cross-sectional regression of average per-asset volatility on the $\log_{10}$ market-cap share, estimated on the 36 non-stablecoin cryptoassets, together with Bitcoin's residual position after

controlling for size. The slope on $\log_{10}$ capitalization is negative and significant in each component, consistent with the size-volatility regularity. After residualisation, the picture separates sharply across components. In *total* volatility, Bitcoin retains the first place out of 36; this is the only result we read as a structural property. In the *common* and *idiosyncratic* components, by contrast, its residual ranks are 7 and 6 out of 36 respectively, in the lower decile but not in the first place; the unconditional gap for these components is therefore partially attributable to capitalization. The reading we adopt is deliberately modest: Bitcoin is the unconditional benchmark of low crypto volatility and the structural benchmark in total volatility once size is controlled for; in the separate components, part of the gap is a size effect.⁹

4.4. Determinants of common and idiosyncratic volatility

The previous three subsections deliver a coherent description of the crypto volatility cross-section: organised in two sharp regimes, structured by capitalization tier and anchored by Bitcoin. What remains is the most diagnostic plane for the research question: whether the channels through which crypto volatility responds to the rest of the economy are the ones that operate in conventional markets, or whether the crypto market is a closed system that responds only to its own internal forces. To examine this, we estimate, for each cryptoasset i and each component $k \in \{c, id\}$, the per-asset regression of Equation 7, with the seven structural variables as standardised regressors and Newey-West HAC standard errors. Table 5 reports the coefficients $\beta_{i,j}^{(k)}$ with significance stars. Panel A presents the common component; Panel B, the idiosyncratic. Coefficients are multiplied by 100, so they read as the average daily change associated with a one-standard-deviation move in the regressor.

Two patterns run through the table. The common component, first of all, is dominated by its own autoregressive structure: once the lagged values of $\sigma_{i,t}^c$ are controlled for, no structural variable enters with economically large or systematically significant coefficients in any of the 38 regressions. Panel A is therefore reported in summary form (mean, median, standard deviation and fraction of cryptoassets significant at conventional levels). The nearly uniform magnitude of the per-asset coefficients in Panel A reflects the factor structure imposed by the decomposition: each asset's common volatility is proportional to its loading on the latent factors, so the regression returns the same factor-level coefficient rescaled by the loading.

The idiosyncratic component, by contrast, admits a more interpretable story and is reported in detail per asset in Panel B. The crypto-native Fear & Greed index is positive and significant in nearly the entire cross-section (at the 5% level for 98% of cryptoassets): greed in crypto-native sentiment consistently amplifies idiosyncratic dispersion. To put magnitudes on an interpretable scale, a one-standard-deviation move in the index lifts Bitcoin's idiosyncratic volatility by roughly 0.32 percentage points per day, about 16% of its average idiosyncratic volatility (1.99%). The effects are larger in absolute terms for the more volatile cryptoassets and stable in proportional terms across the cross-section. The GPR appears as a secondary, less robust channel, with a negative sign and significance at the 5% level for about 28% of cryptoassets, concentrated in mid-caps and a few large-caps; the pattern suggests that, in those assets, geopolitical risk compresses asset-specific dispersion in favour of comovement. The VIX, the EPU, the NFCI, the dollar and

⁹The discrepancy between the first-place position in residualised total volatility and the sixth- and seventh-place ranks in the residualised components reflects the algebraic structure of the exercise: the three regressions are run separately and the residuals are not algebraically tied by $\sigma^{T2} = \sigma^{c2} + \sigma^{id2}$. Online Appendix Table 4 reports the unconditional and residualised ranks side by side for the eight bottom non-stablecoin cryptoassets and explains the pattern.

oil, by contrast, show no incremental explanatory power beyond own persistence.

The selective picture delivered by Panel B admits two possible readings, both examined in the appendix. First, the result might depend on frequency: the EPU, the GPR and the VIX have typically been identified in the crypto literature at weekly or monthly horizons (Demir et al., 2018; Fang et al., 2019; Bouri et al., 2018), so a daily null does not directly contradict that literature. Appendix Table 17 re-estimates the regression for the four mega-caps under weekly aggregation: Fear & Greed retains signs and significance with even larger magnitudes, while the remaining macro variables continue to show no systematic significance at that frequency. The daily reading is therefore preserved at the weekly horizon as well. Second, it could be an artefact of correlation among stress proxies: the NFCI and the VIX are correlated in the sample, and their coexistence in the regression could mask a horse race between them. Appendix Table 16 reports the VIX coefficient in the four mega-caps under the full specification and under an alternative that drops the NFCI: the VIX estimate remains essentially unchanged and insignificant in both, ruling out this explanation. Finally, the moving-block bootstrap inference reported in Section 4.5 to accommodate the generated-regressor structure (Pagan, 1984) preserves the central conclusion.¹⁰

¹⁰Online Appendix Table 3 reports the exhaustive bootstrap over the 76 regressions (38 cryptoassets $\times$ 2 components). Fear & Greed remains significant at the 5% level for 92% of cryptoassets under the bootstrap (98% under HAC); the GPR drops to 0% under the bootstrap (28% under HAC), confirming that it is Fear & Greed, not the GPR, that is the robust channel. The VIX, the EPU, the NFCI, the dollar and oil remain at 0% under either method.

Table 5: Determinants of common and idiosyncratic volatility: per-asset OLS with HAC standard errors.

Asset	VIX	EPU	Fear&Greed	Fin. Cond.	Geo. Risk	USD Ret.	Oil Ret.
Panel A: Common (factor) volatility (cross-sectional summary, $N = 38$ regressions)							
Mean coefficient	0.017	0.002	-0.001	-0.004	-0.005	-0.008	-0.017
Median coefficient	0.018	0.003	-0.001	-0.004	-0.006	-0.009	-0.019
Coef. std. dev.	0.005	0.001	0.000	0.001	0.001	0.002	0.005
% sig. at 5%	0%	0%	0%	0%	0%	0%	0%
% sig. at 1%	0%	0%	0%	0%	0%	0%	0%
Panel B: Idiosyncratic volatility							
<i>Mega-caps</i>							
Bitcoin	0.024	-0.050	0.315***	-0.019	-0.133	0.009	0.029
Ethereum	0.051	-0.051	0.244***	-0.046	-0.156**	-0.027	0.002
BNB	0.015	-0.083	0.545***	-0.058	-0.165*	-0.042	0.025
XRP	-0.003	-0.093	0.523***	0.005	-0.181	-0.026	0.073
<i>Large-caps</i>							
Cardano	0.017	-0.093	0.503***	-0.133	-0.190	-0.059	-0.019
Dogecoin	-0.013	-0.131	0.612***	-0.049	-0.140	-0.021	0.083
TRON	-0.002	-0.050	0.288**	-0.003	-0.207**	-0.008	0.074
Chainlink	0.116	-0.112	0.372**	-0.111	-0.249**	0.004	0.038
BCH	0.089	0.002	0.482***	-0.081	-0.042	-0.077	0.054
Litecoin	0.048	-0.049	0.501***	-0.067	-0.148*	-0.068	0.046
Stellar	-0.017	-0.075	0.443***	0.006	-0.199**	-0.016	0.131*
Filecoin	-0.004	-0.150	0.762***	-0.110	-0.353**	-0.029	0.080
Monero	-0.009	-0.083	0.357***	-0.037	-0.056	-0.024	0.031
ETC	0.034	-0.045	0.237**	-0.121	-0.094	-0.092	0.028
<i>Mid-caps</i>							
Maker	0.070	-0.007	0.491***	-0.041	-0.108	-0.079	0.083

Continued on next page

Table 5 (continued)

Asset	VIX	EPU	Fear&Greed	Fin. Cond.	Geo. Risk	USD Ret.	Oil Ret.
Tezos	0.024	-0.088	0.247**	-0.022	-0.220**	0.007	0.057
EOS	0.107	-0.068	0.261**	-0.069	-0.208**	-0.063	-0.006
Theta	0.092	-0.113	0.637***	-0.066	-0.231	0.008	0.069
VeChain	0.043	-0.091	0.575***	-0.097	-0.163	-0.031	0.048
Synthetix	0.031	-0.118	0.572***	-0.128	-0.212	-0.039	0.037
Decentraland	0.037	-0.085	0.531***	0.050	-0.382**	0.036	0.078
Quant	0.122	-0.110	0.248	-0.130	-0.353*	-0.034	0.043
Nexo	0.023	-0.065	0.405***	-0.016	-0.337**	-0.008	0.069
Enjin	0.229	-0.078	0.613***	-0.019	-0.234	0.019	0.099
Zcash	0.041	-0.010	0.276**	-0.004	0.019	-0.069	0.045
OMG	-0.086	-0.157	0.693***	-0.070	-0.271*	-0.010	0.049
Dash	0.041	-0.047	0.329***	-0.021	-0.159	-0.052	0.028
BAT	0.212	-0.014	0.345***	-0.014	-0.189	-0.013	0.032
IOTA	0.013	-0.088	0.360***	-0.057	-0.256**	-0.012	0.036
Neo	-0.047	-0.112	0.392***	-0.114	0.050	0.023	-0.047
Waves	0.016	-0.106	0.491***	-0.131	0.040	-0.054	0.088
Qtum	0.081	-0.004	0.458***	-0.066	-0.078	-0.042	0.039
Zilliqa	0.002	-0.068	0.473***	-0.057	-0.115	-0.083	0.034
Ontology	0.011	-0.024	0.423***	0.010	-0.108	-0.046	0.037
ICON	0.117	-0.021	0.409***	-0.055	-0.181	-0.115	0.067
Golem	0.007	-0.092	0.584***	0.007	-0.052	-0.041	0.082
<i>Stablecoins</i>							
USDT	0.002	-0.002	0.010***	-0.001	-0.005**	-0.001	0.002
USDC	-0.001	-0.000	0.005**	-0.002	-0.001	0.001	-0.002

Note: Per-asset OLS regressions of the corresponding volatility component on standardised structural market variables (zero mean, unit variance). Each regression includes a constant and lags 1, 5 and 21 of the dependent variable as autoregressive controls (coefficients suppressed). Newey-West (1987) HAC standard errors with $L = 7$ lags. **All reported coefficients are multiplied by 100** to facilitate readability; they represent the average daily change in $\sigma_{i,t}^c$ associated with a one-standard-deviation move in the regressor. The stablecoins (USDT, USDC) are reported in a separate sub-panel; their volatility is small but non-zero. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.5. Robustness

The four results above rest on identification choices that the COVOL decomposition embeds. They remain to be probed. Appendix 2.6 reports a battery of exercises designed to examine each choice independently; the results are summarised here and pointers to the appendix detail are provided when relevant.

The first probe concerns the regime architecture. The 30% and 55% thresholds used in Section 4.1 are, it should be recalled, descriptive labels, not econometrically estimated boundaries. A Markov-switching regression in the spirit of Hamilton (1989) on the 21-day centred moving average of MS_t identifies a low state with endogenous mean 31.5% and a high state with endogenous mean 49.2%, persistence probabilities of 0.98 and 0.99, and inferred durations of 64 and 70 days (Appendix Table 6). The endogenous means are not numerically identical to the descriptive thresholds (the high state is some 6 percentage points below 55%), but the existence of two well-separated and persistent regimes is confirmed: the smoothed marginal probabilities exceed 0.9 or fall below 0.1 for 94% of the sample. The crypto market thus spends almost all of its time decisively in one of two states, not in a continuum of intermediate states. Figure 2 plots the smoothed probabilities.

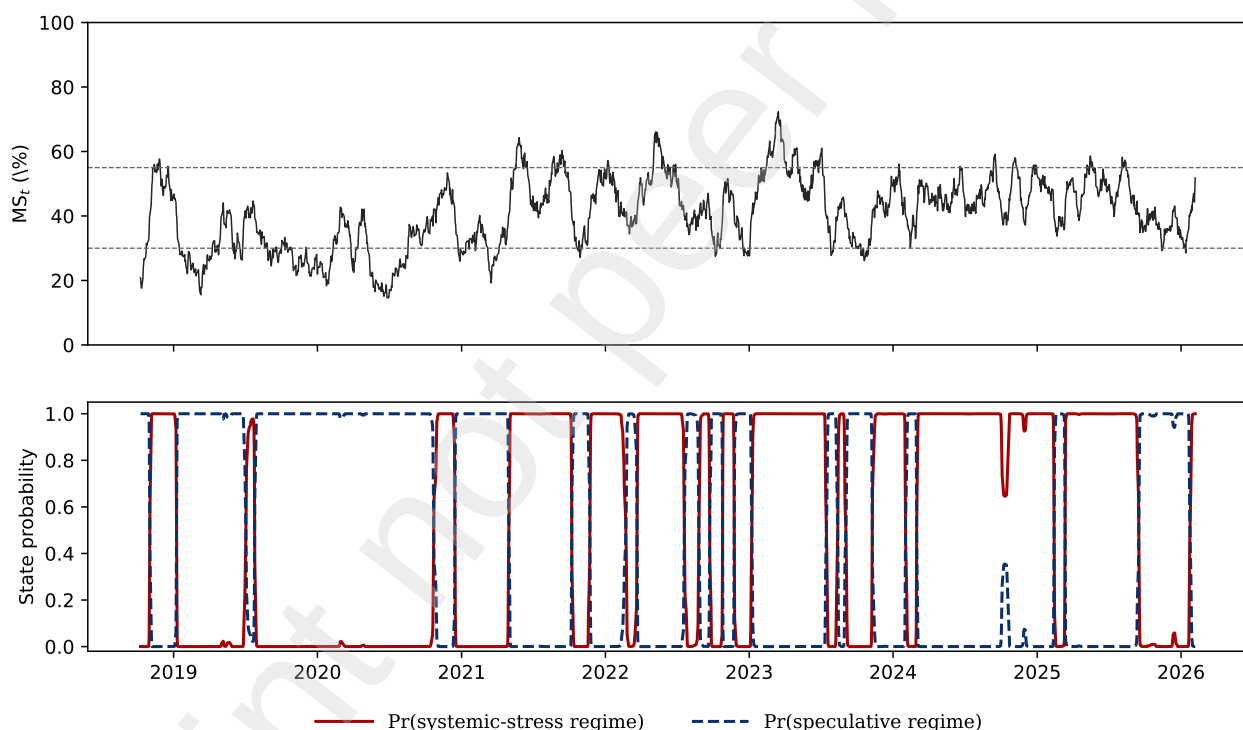


Figure 2: Markov-switching regime identification. Upper panel: daily MS_t in percent, with horizontal lines at 30% and 55%. Lower panel: smoothed probabilities of the systemic-stress and speculative regimes, from the two-state Markov-switching specification reported in Appendix Table 6.

The raw multipliers of 26 and 10 times that Table 2 reports for the SEC v. Ripple and BTC ETF windows raise, within the same regime plane, a separate question: are they sustained common-volatility episodes or single-day shocks embedded in quiet windows? The sensitivity of the trimmed mean (Appendix Table 7), which drops the highest φ_t day and recomputes the multiplier, settles the matter. The SEC v. Ripple multiplier falls from $26.7\times$ to $0.3\times$ the sample mean; the BTC ETF multiplier falls from $9.5\times$ to $0.4\times$. The LUNA / Terra window, by contrast, retains a $1.1\times$ multiplier after trimming, the only case in which the trimmed multiplier remains above one. The

COVOL framework identifies φ_t from the second cross-sectional moment of standardised returns, so the daily estimate is correctly specified for single-day shocks; what is fragile is the within-window arithmetic mean as a summary of multi-week episodes, dominated as it is by the heavy tail of the daily distribution. The moving-block bootstrap confidence intervals for the event multiplier (Appendix Table 10) are wide for the same reason: the bootstrap reflects the high probability of drawing a block without the extreme day. The ranking of the episodes, however, is preserved over symmetric windows of ± 5 , ± 10 and ± 20 days (Appendix Table 11).

The decomposition itself is also robust to the form of cross-sectional aggregation. A leave-one-out aggregate of $\sigma_{i,t}^c$ that drops XRP, Bitcoin and Ethereum in turn remains essentially unchanged during the SEC v. Ripple and BTC ETF windows (Appendix Table 12), indicating that the elevated common volatility is shared across the cross-section and is not a mechanical consequence of a single high-loading asset. Reaggregation without stablecoins (Appendix Table 13) preserves the episode ranking and the regime structure; ADF and KPSS tests on the aggregate series are jointly consistent with stationarity (Appendix Table 15); and the mean of MS_t remains in 38–43% across pre/post-2022 and pre/post-ETF subsamples (Appendix Table 14).

A separate concern, this one purely statistical, is inference. The volatility series used in Equation 7 are themselves estimated quantities, so the asymptotic HAC standard errors understate sample uncertainty (Pagan, 1984). Moving-block bootstrap standard errors for the regressions of the four mega-caps (Appendix Table 3) are larger than their HAC counterparts but preserve the qualitative conclusion: Fear & Greed remains significant in the four mega-caps, while the VIX, the EPU, the NFCI, the GPR, the dollar and oil remain insignificant.

More substantive, however, is the question of survivorship bias. The universe of 38 assets is conditioned on continuous trading through to the end of the sample, which excludes Terra / LUNA, TerraUSD and FTT. To bound the magnitude of the bias, we reaggregate the cross-section by including those three assets up to their last trading dates (LUNA and TerraUSD through 11 and 13 May 2022, and FTT through 30 November 2022). For each failed token we fit a univariate sGARCH(1,1)-Student- t and treat the conditional variance as fully idiosyncratic, an attribution that is the worst case for the paper’s answer to the research question, since it maximally inflates the cross-sectional idiosyncratic sum and pushes the market share maximally downward. Appendix Table 8 reports the comparison. During the LUNA / Terra window, including LUNA and TerraUSD lowers the average market share by 2.0 percentage points (from 76.4% to 74.4%) and the median by 1.2 points: the systemic-stress classification remains intact. During the FTX window, in contrast, including FTT lowers the average market share by 13.9 points (from 77.6% to 63.7%) and the median by 17.2 points: the systemic-stress classification is preserved (mean and median above 55%), but the magnitude is significantly overstated by the survivorship-conditioned panel. The qualitative ranking of episodes survives the inclusion of the failed tokens; the magnitudes during FTX-type windows, on the other hand, should be read as upper bounds on the systemic content.

What remains is the most direct probe for the research question, the one that is comparative in nature: how does the crypto cross-section compare with a non-crypto benchmark estimated under the same procedure? We apply the same APC + COVOL machinery to two equity benchmarks over the same calendar window. The primary one is a panel of 38 individual large-cap stocks from the S&P 500, selected by ranking at the end of 2025 and dimensionally comparable to the crypto panel ($N = 38$, individual assets). The secondary one is the panel of 11 SPDR sector ETFs (XLK, XLF, XLE, XLV, XLY, XLP, XLI, XLB, XLU, XLRE, XLC), which summarises the broad factor at the sector level. The two benchmarks bracket the comparison: the first matches in dimensionality and individual-asset structure; the second reduces idiosyncratic noise through

sector aggregation. Both deliver the same qualitative reading. The unconditional distribution of MS_t in the individual-stock panel has mean 35.0% (against 41.1% for crypto), median 30.8% (38.8%), interdecile range [1.2%, 75.3%] ([3.0%, 83.5%]) and standard deviation 27.7% (29.7%). In the SPDR panel the mean is 36.0% and the median 32.9%, both within one percentage point of the individual-stock figures. The Markov-switching specification of Hamilton (1989) identifies a low state in the 28.3–31.5% range and a high state in the 43.4–49.2% range across the three cross-sections, with high persistence probabilities and 60–70-day durations on the crypto panel. The low regime is essentially identical between crypto, individual stocks and sector ETFs (Appendix Tables 11–12). At the episode level, the contrast is informative: during the COVID-19 selloff, the only universal stress event in the sample, crypto and equities report nearly identical $\overline{MS}$; during crypto-native events such as LUNA, the USDC depeg and the BTC ETF rally, the crypto $\overline{MS}$ exceeds the equity counterpart by 11 to 18 percentage points (Appendix Table 18). That both panels deliver the same picture rules out that the comparison reflects the diversification embedded in the ETFs rather than a structural similarity. The comparative answer is, therefore, the one the title of the paper announces: the crypto regime architecture is recognisable relative to the equity one in unconditional terms, whether against individual stocks or sector portfolios; the differential operates in the upper tail of stress and concentrates on crypto-specific events.

5. Conclusions

We investigate whether the volatility dynamics of cryptocurrencies are governed by the same regularities as conventional markets or whether they constitute a genuinely sui generis structure. To make the question operational, we apply to a contemporary crypto panel the asymptotic principal components factor model of Connor et al. (2006), implemented via the iterative algorithm of Jones (2001), together with the residual COVOL framework of Engle & Campos-Martins (2023). The procedure delivers, for each asset and each day, a decomposition of total volatility into a common cross-sectional component and an asset-specific residual, on 38 cryptoassets that account for about 91% of market capitalization as of December 2025 and are observed daily between October 2018 and February 2026.

The evidence that emerges accumulates in the same direction. The crypto cross-section alternates between two clearly distinct regimes: a systemic-stress regime in which the latent common factor accounts for more than 55% of the cross-sectional volatility, and a speculative regime in which idiosyncratic dynamics dominate and the common market share falls below 30%. An endogenous Markov-switching identification (Hamilton, 1989) corroborates that architecture, with regime means at 31% and 49% and persistence probabilities close to 0.98 in each state. Within the architecture, the two largest single-day cross-sectional shocks in the entire sample are recorded at the SEC v. Ripple ruling and the spot Bitcoin ETF approval of 2023, but the surrounding windows sit at levels close to the sample median: a trimmed mean that drops the highest day reduces the SEC v. Ripple and BTC ETF multipliers from $26\times$ and $10\times$ to $0.31\times$ and $0.37\times$ the sample mean. LUNA / Terra stands as the only episode in the period with sustained elevation of common volatility over a multi-week window, and therefore the only case of sustained systemic stress that the decomposition identifies.

At the individual-asset level, Bitcoin sits at the lower extreme of the crypto volatility distribution in absolute terms and retains the first place in total volatility after controlling for capitalization, which we read as a structural property of its anchor role; in the separate components, by contrast, a non-trivial portion of the gap reflects the size effect. Macro-financial transmission at the daily frequency turns out, in turn, to be markedly selective: crypto-native sentiment, captured

by the Fear & Greed index, amplifies the idiosyncratic volatility of nearly the entire cross-section, while the VIX, the EPU of Baker et al. (2016), the GPR of Caldara & Iacoviello (2022), the NFCI, the dollar and oil add no incremental explanatory power. The result is preserved at the weekly horizon, where Fear & Greed retains magnitudes and significance, ruling out that the selectivity observed at the daily frequency is an artefact of noise or of frequency mismatch relative to Demir et al. (2018), Fang et al. (2019) and Bouri et al. (2018).

These findings speak to several established literatures. The cross-sectional factor evidence of Liu et al. (2022) on crypto *returns* finds its counterpart here on *volatility*, with the addition that the factor structure relevant for risk management is dynamic rather than time-invariant. The univariate GARCH evidence (Dyhrberg, 2016; Katsiampa, 2017) is complemented by an explicit decomposition that separates the common from the asset-specific. The literature on stress episodes (Briola et al., 2023; Conlon et al., 2024; Galati & Capalbo, 2024) is enriched with the documentation of the regulatory and ETF-related spikes of 2023. The qualitative debate about Bitcoin (Baur et al., 2018; Härdle et al., 2020) receives a quantitative anchor that explicitly recognises the role of capitalization. And the existing COVOL literature (Engle & Campos-Martins, 2023; Campos-Martins & Hendry, 2024; Pham et al., 2025, 2026) is extended from the sectoral to the individual-asset level on a contemporary crypto universe.

The comparative dimension of the question, anchored by two equity benchmarks estimated under the same procedure, delivers the answer that the title of the paper announces. Both the panel of 38 individual large-cap stocks from the S&P 500 and the panel of 11 SPDR sector ETFs deliver the same picture: the unconditional distribution of the daily common market share, the endogenous Markov-switching boundaries (low state at 28–32% and high state at 43–49% across the three panels) and the response during the COVID-19 selloff are essentially indistinguishable between crypto and equities. The differential operates in the upper tail of stress and concentrates on crypto-specific events such as LUNA, the USDC depeg and the BTC ETF anticipation rally, where the crypto market share exceeds the equity counterpart by 11 to 18 percentage points. That both panels deliver the same picture rules out that the comparison reflects the diversification embedded in the ETFs rather than a structural similarity. The crypto market, in the space of volatility, is continuous with the equity benchmark in unconditional terms, and what appears as genuinely different is narrower than the popular narrative suggests: it concentrates on the magnitude of crypto-native shocks in the upper tail.

The practical implications cut across investment, regulation and risk management. For investors, the regime structure implies that the diversification benefits within the crypto market are state-contingent and are lost during systemic-stress episodes; intra-crypto diversification is therefore a poor substitute for hedging through assets outside the crypto factor structure. For regulators, the concentration of the largest cross-sectional spikes on regulatory and institutional events suggests that interventions presented as narrow, namely a ruling on a single cryptoasset or the approval of an investment vehicle, can produce single-day systemic footprints not seen during the most heavily covered crypto-native failures; the design and communication of such interventions therefore carry systemic implications that should be internalised in the regulatory calendar. For risk managers, the insulation of crypto volatility from the VIX, the EPU, the NFCI, the GPR, the dollar and oil at the daily frequency implies that crypto exposures cannot be hedged with conventional macro proxies; crypto-specific risk budgets must be built and monitored on their own terms.

The robustness exercises in Section 4.5 support, taken together, each of these findings: the regime architecture is corroborated by the endogenous Markov-switching identification; the multiplicative magnitudes of the 2023 episodes are accompanied by leave-one-out aggregates and boot-

strap intervals; the macro regressions are preserved by conservative inference under generated regressors (Pagan, 1984); the comparative dimension is anchored by the sector equity benchmark; and the survivorship bias is mitigated by reaggregating the cross-section with LUNA, TerraUSD and FTT included up to their last trading dates, an exercise under which the qualitative regime classification is preserved, although the systemic magnitude during the FTX window is significantly reduced. Two substantive limitations remain, nonetheless, worth flagging. One is the choice of univariate model: the strategy assumes an sGARCH(1,1)-Student-t specification for each cryptoasset, and an extension to EGARCH or GJR-GARCH would assess whether the findings are robust to asymmetric responses to good and bad news. The other concerns the universe: the augmented exercise with failed tokens treats their full variance as idiosyncratic and adds only three tokens, while a dynamically rebalanced universe in the spirit of Liu et al. (2022), in which the panel composition varies with t , would deliver a more complete treatment of survivorship bias.

Several avenues for future research follow naturally from these limitations and from the central findings. The decomposition can be extended to higher-frequency data to study the intraday propagation of regulatory and institutional shocks; combined with directional asymmetries in the spirit of Pham et al. (2026) to separate the response of the common factor to good and bad news; linked with on-chain measures of network activity, holder concentration and stablecoin flows to identify the structural microfoundations of regime changes; integrated with the role of stablecoins in the transmission of volatility between crypto and currency markets; and applied to the post-2024 institutionalisation of the crypto market, particularly the gradual entry of regulated investment vehicles, which is likely to shift the balance between common and idiosyncratic volatility going forward.

Cryptocurrencies have been described, since their inception, as a financially distinctive asset class. The evidence assembled here answers the research question with clarity: that characterisation is overstated when read through the lens of cross-sectional volatility. The crypto market exhibits a regime architecture that is quantitatively comparable to that of an equity benchmark estimated under the same procedure, a hierarchy in which institutional and regulatory events weigh more than purely technological shocks, a reference asset that anchors the cross-section, and a selectivity of the macro-financial channels through which volatility is transmitted. What appears as genuinely different is narrower than the popular narrative suggests, and concentrates on the magnitude of crypto-native shocks in the upper tail. In the space of volatility, this time is not so different.

6. Appendix

A.1 Eigenvalue scree plot

Figure 3 reports the share of variance captured by each principal component extracted from the panel of standardised crypto log-returns. The first factor explains 56.2% of the total cross-sectional variance, the second adds 4.1% and the third 2.6%. The eigenvalues flatten from the third component onward, supporting the parsimonious choice $K = 2$ adopted in the main analysis on the basis of the Bai-Ng information criterion.

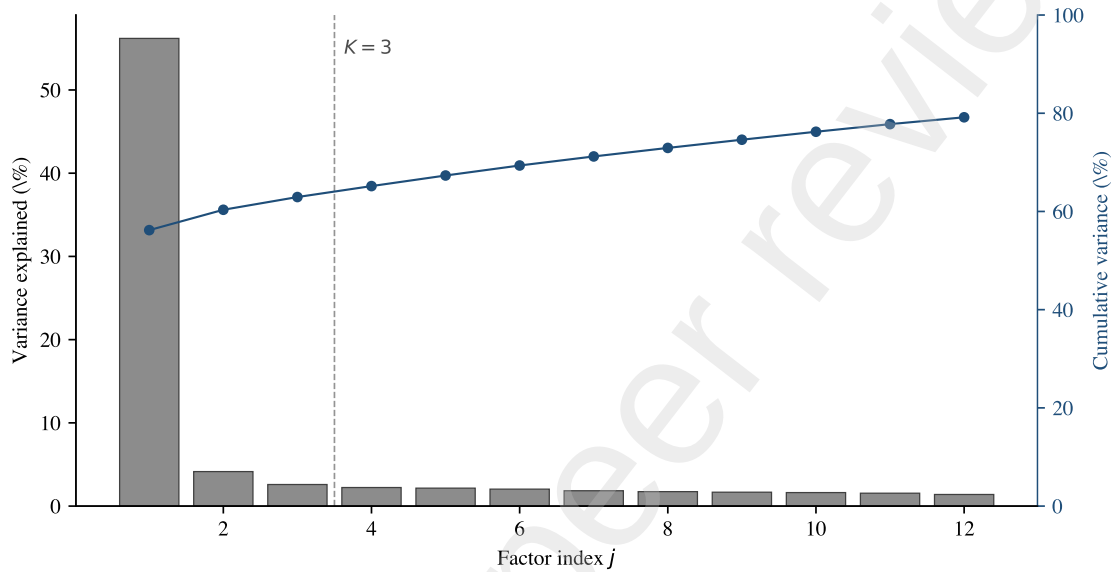


Figure 3: Eigenvalue scree plot of the standardised return panel. Bars show the percentage of total cross-sectional variance explained by each principal component; the line (right axis) shows the cumulative variance.

The tables and figures supporting additional robustness exercises and detailed per-asset diagnostics have been moved to the Online Appendix to keep the main document compact. The Online Appendix includes: (i) the reconciliation of the two cross-sectional aggregations of the common market share; (ii) a single-factor robustness on the non-stablecoin cross-section; (iii) the exhaustive moving-block bootstrap inference for the 76 idiosyncratic-volatility regressions; (iv) the algebraic decomposition of Bitcoin's first-place position in total volatility; (v) the per-asset descriptive statistics of the volatility components and of the raw daily returns; (vi) the per-asset pairwise volatility differentials against Bitcoin; (vii) the full per-asset universe table; (viii) the factor loadings under the alternative $K = 3$ specification; and (ix) the auxiliary robustness tables (event-window bootstrap intervals, robustness to symmetric windows, leave-one-out aggregates, stablecoin-free aggregates, subsample stability, stationarity tests, NFCI/VIX horse race, weekly-frequency macro regressions, Bitcoin effect-size diagnostics, residualisation of Bitcoin's position, the SPDR sector equity benchmark detail and the equity time-series figure).

Table 6: Markov-switching regime identification on the daily common share MS_t .

Model	Mean lo (%)	Mean hi (%)	SD lo (%)	SD hi (%)	$P_{lo,lo}$	$P_{hi,hi}$	Dur lo (days)	Dur hi (days)	AIC	BIC
MS(2)	31.5	49.2	6.3	6.1	0.984	0.986	63.7	70.4	12605	12638
MS(3)	28.4	52.3	5.0	5.0	—	—	39.5	34.1	11511	11577

Note: Markov-switching regression of the 21-day centred rolling mean of MS_t (in percent) on a constant with regime-specific mean and variance. The 21-day input window is chosen to identify sustained regimes; shorter windows produce excessively frequent transitions, while longer windows are degenerate. The endogenously estimated low-MS state (speculative regime) and high-MS state (systemic-stress regime) match the 30%/55% qualitative thresholds discussed in the text. $P_{lo,lo}$ and $P_{hi,hi}$ are the daily transition probabilities of remaining in the low and high state, respectively. The Bayesian information criterion is reported as a model-selection diagnostic. Sample: $T = 1,908$ daily observations.

Table 7: Sensitivity of the event-window multiplier $\bar{\varphi}_{event}/\bar{\varphi}_{full}$ to the single largest day within each window.

Episode	N	Mean of φ_t		φ_{max}	Multiplier	
		Raw	Trimmed		Raw	Trimmed
SEC v Ripple ruling	16	0.1012	0.0012	1.601	26.7×	0.3×
BTC ETF anticipation	44	0.0361	0.0014	1.526	9.5×	0.4×
COVID crash	32	0.0107	0.0029	0.253	2.8×	0.8×
USDC depeg / SVB	9	0.0093	0.0030	0.060	2.5×	0.8×
China mining ban	55	0.0060	0.0032	0.161	1.6×	0.8×
LUNA / Terra	33	0.0056	0.0042	0.049	1.5×	1.1×
Altcoin season	41	0.0043	0.0038	0.024	1.1×	1.0×
Trump election rally	30	0.0035	0.0026	0.030	0.9×	0.7×
FTX collapse	33	0.0030	0.0021	0.032	0.8×	0.5×
BTC halving 2024	33	0.0024	0.0018	0.020	0.6×	0.5×
Bull-run peak	45	0.0020	0.0019	0.007	0.5×	0.5×
Nov 2021 ATH	33	0.0018	0.0014	0.012	0.5×	0.4×
DeFi summer	66	0.0009	0.0008	0.007	0.2×	0.2×

Note: For each event window, “Raw” is the arithmetic mean of φ_t over the N trading days of the window, and “Trimmed” is the same mean after dropping the single trading day with the largest in-window φ_t . The multiplier divides each mean by the full-sample mean ($\bar{\varphi}_{full} = 0.0038$).

$varphi_{max}$ is the largest single-day value of

$varphi_t$ within the window. Episodes are sorted by the raw multiplier in descending order. The contrast between the raw and the trimmed multipliers identifies which episodes are characterised by a sustained elevation of

$varphi_t$ (small change between Raw and Trimmed) and which are driven by a single extreme day (Raw

gg Trimmed).

Table 8: Survivorship-bias robustness: episode aggregates with failed tokens included up to their last available trading date.

Episode	Failed tokens added	38-asset universe		Augmented universe		$\Delta \overline{MS}$ (pp)
		$\overline{MS}$	$\widetilde{MS}$	$\overline{MS}$	$\widetilde{MS}$	
LUNA / Terra	LUNA, UST	76.4%	80.2%	74.4%	79.0%	-2.0
FTX collapse	FTT	77.6%	82.5%	63.7%	65.3%	-13.8

Note: Survivorship-bias robustness exercise on the LUNA / Terra and FTX collapse windows. The 38-asset universe reproduces the cross-sectional aggregate of $\sigma_{i,t}^c$ and $\sigma_{i,t}^{id}$ used in Sections 4.1 and 4.5. The augmented universe adds LUNA and TerraUSD (USTC) for the LUNA / Terra window (until their last trading date of 2022-05-11 / 2022-05-13) and FTT for the FTX window (until 2022-11-30). The failed tokens contribute zero to the common-factor cross-sectional sum (consistent with their negligible loading on the broad market factor) and their entire univariate sGARCH(1,1)-Student- t conditional variance to the idiosyncratic cross-sectional sum (the worst-case attribution for the paper’s central claim, since this maximally inflates γ and pushes MS downward). $\overline{MS}$ and $\widetilde{MS}$ denote the within-window mean and median of the daily common share. $\Delta \overline{MS}$ is the change in the mean common share when the failed tokens are included.

Table 9: Bitcoin as low-volatility anchor: effect-size diagnostics by capitalization tier.

Tier	N	Fraction of days $\sigma_{BTC}^k < \sigma_j^k$			Cohen’s d (σ^T)
		Common	Idio.	Total	
Mega-caps	3	1.00	0.81	0.97	1.11
Large-caps	10	1.00	0.90	0.99	1.00
Mid-caps	22	1.00	0.88	1.00	1.26
All non-stable	35	1.00	0.88	0.99	1.17

Note: Effect-size diagnostics for the position of Bitcoin in the cryptocurrency volatility cross-section. Columns 3–5 report, for each component $k \in \{\text{common, idiosyncratic, total}\}$, the share of trading days on which Bitcoin’s volatility is strictly smaller than that of the corresponding asset, averaged across the cryptoassets within each tier. The last column reports Cohen’s d on the daily differential $\sigma_{j,t}^T - \sigma_{BTC,t}^T$; positive values indicate that Bitcoin sits below the corresponding asset in total volatility. All values are averaged within each tier across the 36 non-stablecoin cryptoassets.

Table 10: Bitcoin’s position in the volatility cross-section after controlling for capitalization.

Component	$\hat{\beta}_{\log \text{ mcap}}$	p -value	BTC raw	BTC resid.	BTC rank
$\bar{\sigma}_i^c$	−0.0033 (0.0005)	0.000	0.0255	−0.0038	7/36
$\bar{\sigma}_i^{\text{id}}$	−0.0022 (0.0010)	0.030	0.0196	−0.0062	6/36
$\bar{\sigma}_i^T$	−0.0042 (0.0007)	0.000	0.0333	−0.0073	1/36

Note: Cross-sectional OLS regression of the per-asset mean volatility component on $\log_{10}$ of the market-capitalization share, estimated over the 36 non-stablecoin cryptoassets. The column $\hat{\beta}_{\log \text{ mcap}}$ reports the slope coefficient with the standard error in parentheses. The slope on $\log_{10}$ market cap is negative and significant for every component, consistent with the size-volatility regularity documented in conventional asset markets. “BTC raw” is the unconditional sample mean of the corresponding component for Bitcoin; “BTC resid.” is the regression residual for Bitcoin; “BTC rank” is the position of the Bitcoin residual within the 36-asset cross-section, ordered from smallest to largest. A rank close to 1 indicates that Bitcoin remains at the bottom of the cross-section even after controlling for size; a rank near the median indicates that the unconditional anchor result is largely a size effect.

Table 11: Distribution of the daily common share MS_t : cryptocurrency cross-section vs panel of 38 individual large-cap S&P 500 stocks.

Statistic	Cryptocurrency MS_t (%)	Equity MS_t (%)	Difference (pp)
P5	1.1%	0.2%	-0.9
P10	3.0%	1.2%	-1.7
P25	13.0%	8.7%	-4.4
P50	38.8%	30.8%	-8.0
P75	67.0%	58.5%	-8.5
P90	83.5%	75.3%	-8.2
P95	89.8%	82.9%	-6.9
Mean	41.1%	35.0%	-6.2
Std. dev.	29.7%	27.7%	-2.0

Note: Distribution of the daily common share MS_t over the 1838 trading days for which both panels are observed. The cryptocurrency series is computed on the 38-asset universe of the main analysis (Section 4.1); the equity series is computed on a panel of 38 individual large-cap S&P 500 stocks (selected by market capitalization rank as of late 2025) using an analogous APC + COVOL pipeline with $K = 1$ latent factor. The dimensionality of the two cross-sections is matched at $N = 38$, and each unit is an individual asset rather than a diversified portfolio. Quantiles are reported in percent. The right-most column reports the difference (equity minus cryptocurrency) in percentage points.

Table 12: Markov-switching identification of regimes on the daily common share MS_t across cross-sections.

Cross-section	Mean low (%)	Mean high (%)	SD low (%)	SD high (%)	Dur low (days)	Dur high (days)
Cryptocurrency (38 assets)	31.5	49.2	6.3	6.1	63.7	70.4
Equity individual stocks ($N=38$)	28.3	43.4	6.3	6.8	13.7	10.7
Equity SPDR sectors ($N=11$)	29.0	44.4	6.7	6.7	12.8	10.7

Note: Two-state Markov-switching regression of MS_t for each cross-section, with regime-specific means and variances. The cryptocurrency row uses the 21-day rolling-mean input of Table 6; the equity rows use the seven-day rolling-mean input of the equity benchmark scripts. Differences in inferred regime durations across rows reflect this difference in input smoothing. The cryptocurrency row reproduces the estimates of Table 6. The individual-stock row reports the analogous estimates on the 38-stock S&P 500 panel; the SPDR-sector row reports the estimates on the 11-ETF panel discussed in Online Appendix Table 20. The individual-stock benchmark is dimensionally matched to the cryptocurrency cross-section ($N = 38$) and consists of individual assets rather than diversified portfolios; the SPDR-sector panel summarises the broad market factor at the sector level.

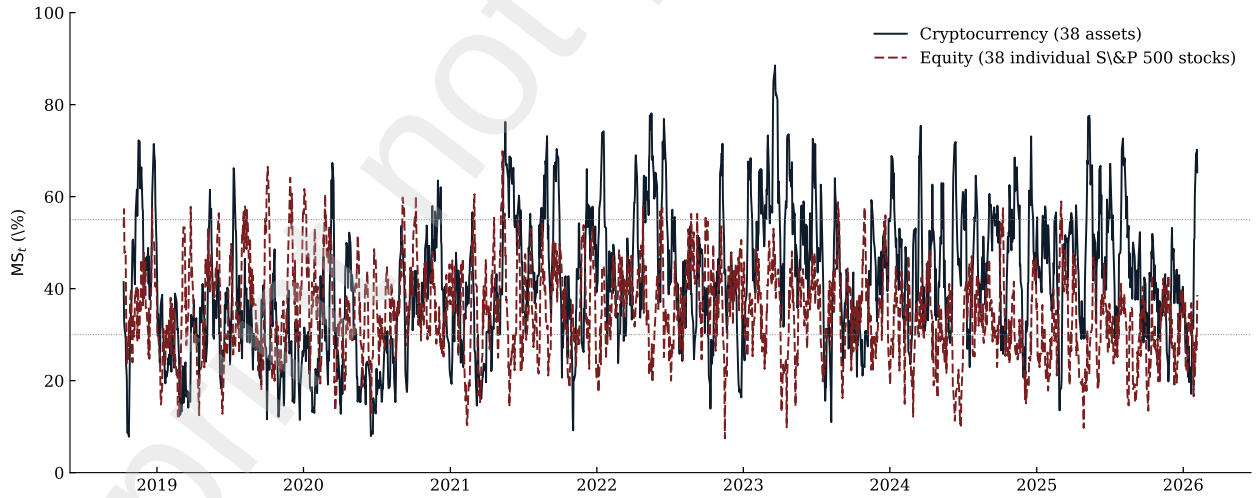


Figure 4: Daily common market share MS_t for the crypto cross-section (38 cryptoassets, dark-blue solid line) and for an equity benchmark of 38 individual large-cap stocks from the S&P 500 (dark-red dashed line), seven-day centred moving averages in percent. The horizontal dotted lines at 30% and 55% mark the threshold values used in Section 4.1.

A.2 Markov-switching regime identification

A.3 Event-window robustness: trimmed multipliers and survivorship

A.4 Bitcoin's position in the cross-section

A.5 Equity benchmark: individual S&P 500 stocks

References

- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723. <https://doi.org/10.1109/TAC.1974.1100705>.
- Akyildirim, E., Aysan, A. F., Cepni, O., & Darendeli, S. P. C. (2024). The FTX collapse and cryptocurrency market dynamics. *Finance Research Letters*, 59, 104796. <https://doi.org/10.1016/j.frl.2023.104796>.
- Aramonte, S., Huang, W., & Schrimpf, A. (2021). DeFi risks and the decentralisation illusion. *BIS Quarterly Review*, (pp. 21–36).
- Bai, J. & Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221. <https://doi.org/10.1111/1468-0262.00273>.
- Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *The Quarterly Journal of Economics*, 131(4), 1593–1636. <https://doi.org/10.1093/qje/qjw024>.
- Barigozzi, M. & Hallin, M. (2017). Generalized dynamic factor models and volatilities: Estimation and forecasting. *Journal of Econometrics*, 201(2), 307–321. <https://doi.org/10.1016/j.jeconom.2017.08.010>.
- Baur, D. G., Hong, K., & Lee, A. D. (2018). Bitcoin: Medium of exchange or speculative assets? *Journal of International Financial Markets, Institutions and Money*, 54, 177–189. <https://doi.org/10.1016/j.intfin.2017.12.004>.
- Bergsli, L. Ø., Lien, A. F., Magdziarz, M., & Westgaard, S. (2022). Forecasting volatility of Bitcoin. *Research in International Business and Finance*, 59, 101540. <https://doi.org/10.1016/j.ribaf.2021.101540>.
- Biais, B., Bisière, C., Bouvard, M., Casamatta, C., & Menkveld, A. J. (2023). Equilibrium Bitcoin pricing. *The Journal of Finance*, 78(2), 967–1014. <https://doi.org/10.1111/jofi.13206>.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1).
- Bouri, E., Das, M., Gupta, R., & Roubaud, D. (2018). Spillovers between Bitcoin and other assets during bear and bull markets. *Applied Economics*, 50(55), 5935–5949. <https://doi.org/10.1080/00036846.2018.1488075>.
- Briola, A., Vidal-Tomás, D., Wang, Y., & Aste, T. (2023). Anatomy of a stablecoin's failure: The Terra-Luna case. *Finance Research Letters*, 51, 103358. <https://doi.org/10.1016/j.frl.2022.103358>.
- Caldara, D. & Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225. <https://doi.org/10.1257/aer.20191823>.

- Campos-Martins, S. & Hendry, D. F. (2024). Common volatility shocks driven by the global carbon transition. *Journal of Econometrics*, 239(1), 105472. <https://doi.org/10.1016/j.jeconom.2023.04.018>.
- Conlon, T., Corbet, S., & Hou, Y. (2024). Contagion effects of permissionless, worthless cryptocurrency tokens: Evidence from the collapse of FTX. *Journal of International Financial Markets, Institutions and Money*, 91, 101940. <https://doi.org/10.1016/j.intfin.2024.101940>.
- Connor, G., Korajczyk, R. A., & Linton, O. (2006). The common and specific components of dynamic volatility. *Journal of Econometrics*, 132(1), 231–255. <https://doi.org/10.1016/j.jeconom.2005.01.030>.
- Da, Z., Engelberg, J., & Gao, P. (2011). In search of attention. *The Journal of Finance*, 66(5), 1461–1499. <https://doi.org/10.1111/j.1540-6261.2011.01679.x>.
- Demir, E., Gozgor, G., Lau, C. K. M., & Vigne, S. A. (2018). Does economic policy uncertainty predict the Bitcoin returns? An empirical investigation. *Finance Research Letters*, 26, 145–149. <https://doi.org/10.1016/j.frl.2018.01.005>.
- Dyrhberg, A. H. (2016). Bitcoin, gold and the dollar – A GARCH volatility analysis. *Finance Research Letters*, 16, 85–92. <https://doi.org/10.1016/j.frl.2015.10.008>.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007. <https://doi.org/10.2307/1912773>.
- Engle, R. F. & Campos-Martins, S. (2023). What are the events that shake our world? Measuring and hedging global COVOL. *Journal of Financial Economics*, 147(1), 221–242. <https://doi.org/10.1016/j.jfineco.2022.09.009>.
- Fama, E. F. & French, K. R. (1992). The cross-section of expected stock returns. *The Journal of Finance*, 47(2), 427–465. <https://doi.org/10.1111/j.1540-6261.1992.tb04398.x>.
- Fang, L., Bouri, E., Gupta, R., & Roubaud, D. (2019). Does global economic uncertainty matter for the volatility and hedging effectiveness of Bitcoin? *International Review of Financial Analysis*, 61, 29–36. <https://doi.org/10.1016/j.irfa.2018.12.010>.
- Galati, L. & Capalbo, F. (2024). Silicon Valley Bank bankruptcy and stablecoins stability. *International Review of Financial Analysis*, 91, 103001. <https://doi.org/10.1016/j.irfa.2023.103001>.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384. <https://doi.org/10.2307/1912559>.
- Härdle, W. K., Harvey, C. R., & Reule, R. C. G. (2020). Understanding cryptocurrencies. *Journal of Financial Econometrics*, 18(2), 181–208. <https://doi.org/10.1093/jjfinec/nbz033>.
- Herskovic, B., Kelly, B., Lustig, H., & Van Nieuwerburgh, S. (2016). The common factor in idiosyncratic volatility: Quantitative asset pricing implications. *Journal of Financial Economics*, 119(2), 249–283. <https://doi.org/10.1016/j.jfineco.2015.09.010>.
- Jones, C. S. (2001). Extracting factors from heteroskedastic asset returns. *Journal of Financial Economics*, 62(2), 293–325. [https://doi.org/10.1016/S0304-405X\(01\)00078-1](https://doi.org/10.1016/S0304-405X(01)00078-1).

- Katsiampa, P. (2017). Volatility estimation for Bitcoin: A comparison of GARCH models. *Economics Letters*, 158, 3–6. <https://doi.org/10.1016/j.econlet.2017.06.023>.
- Liu, J., Makarov, I., & Schoar, A. (2023). Silicon Valley Bank, the USDC depeg, and the stablecoin–banking nexus. *Working paper*. NBER Working Paper No. 31290.
- Liu, Y. & Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *The Review of Financial Studies*, 34(6), 2689–2727. <https://doi.org/10.1093/rfs/hhaa113>.
- Liu, Y., Tsyvinski, A., & Wu, X. (2022). Common risk factors in cryptocurrency. *The Journal of Finance*, 77(2), 1133–1177. <https://doi.org/10.1111/jofi.13119>.
- Newey, W. K. & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. <https://doi.org/10.2307/1913610>.
- Pagan, A. (1984). Econometric issues in the analysis of regressions with generated regressors. *International Economic Review*, 25(1), 221–247. <https://doi.org/10.2307/2648877>.
- Pham, S. D., Han, W., Do, H. X., & Pham, L. (2025). Exploring common volatility in cryptocurrencies: A subsectoral perspective. Working Paper, SSRN 5373305.
- Pham, S. D., Han, W., Nguyen, T. T. T., Pham, L., & Do, H. X. (2026). Good versus bad COVOL in cryptocurrency markets: A measure of asymmetric common volatility. Working Paper, SSRN 6038623.
- Smales, L. A. (2022). Investor attention in cryptocurrency markets. *International Review of Financial Analysis*, 79, 101972. <https://doi.org/10.1016/j.irfa.2021.101972>.
- Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80–82. <https://doi.org/10.1016/j.econlet.2016.09.019>.
- Xu, D., Hu, Y., Corbet, S., & Goodell, J. W. (2023). Volatility connectedness between global COVOL and major international volatility indices. *Finance Research Letters*, 56, 104112. <https://doi.org/10.1016/j.frl.2023.104112>.
- Zhang, W. & Li, Y. (2020). Is idiosyncratic volatility priced in cryptocurrency markets? *Research in International Business and Finance*, 54, 101252. <https://doi.org/10.1016/j.ribaf.2020.101252>.
- Zhang, Z. & Zhao, R. (2023). Good volatility, bad volatility, and the cross section of cryptocurrency returns. *International Review of Financial Analysis*, 89, 102712. <https://doi.org/10.1016/j.irfa.2023.102712>.